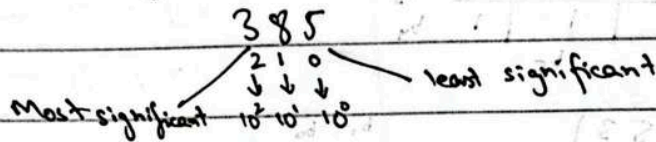


Decimal No. System

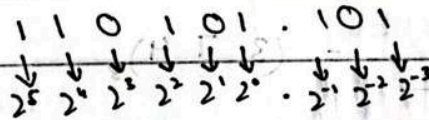
Base 10

Total Digits 10



$$= 3 \times 10^2 + 8 \times 10^1 + 5 \times 10^0 = 300 + 80 + 5 = 385$$

Binary:-



$$= 1 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 + 1 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3}$$

$$= 32 + 16 + 4 + 1 + 0.5 + 0.125 = (53.625)_{10}$$

Octal:-

Base 8: 0-7

eg $(124)_8 = 1 \times 8^2 + 2 \times 8^1 + 4 \times 8^0 = 64 + 16 + 4 = (84)_{10}$

$$(124.31)_8 = 1 \times 8^2 + 2 \times 8^1 + 4 \times 8^0 + 3 \times 8^{-1} + 1 \times 8^{-2} = 84 + 3 \times 8^{-1} + 1 \times 8^{-2} = \dots$$

Hexadecimal:-

Base 16:-

0-9

A $\rightarrow$ 10
B $\rightarrow$ 11
C $\rightarrow$ 12
D $\rightarrow$ 13
E $\rightarrow$ 14
F $\rightarrow$ 15

(3F2)₁₆

$$= 3 \times 16^2 + 15 \times 16^1 + 2 \times 16^0$$

$$= (1010)_{10}$$

(31)₄

$$= 3 \times 4^1 + 1 \times 4^0$$

$$= 12 + 1 = (13)_{10}$$

Convert a number from decimal to any other base:-

To binary

$$(53)_{10} =$$

2	53	
2	26	1
2	13	0
2	6	1
2	3	0
	1	1

$$= (110101)_2$$

$$= 32 + 16 + 4 + 1 = 53$$

To Octal

$$(53)_{10} =$$

8	53	
	6	5

$$= (65)_8$$

$$= 6 \times 8^1 + 5 \times 8^0 = 48 + 5 = 53$$

$$(52)_{10} =$$

8	52	
	4	4
	3	7

$$= (374)_8$$

To Hexadecimal

$$(757)_{10} =$$

16	757	
	47	5
	2	F

$$= (2F5)_{16}$$

$$= 2 \times 16^2 + 15 \times 16^1 + 5 \times 16^0$$

$$= (757)_{10}$$

Decimal Fraction to Binary/Octal/Hexadecimal

$$(0.8125)_{10}$$

Number	Multiply by 2	Integer
0.8125	$2 \times 0.8125 = 1.625$	1
0.625	$2 \times 0.625 = 1.25$	1
0.25	$2 \times 0.25 = 0.5$	0
0.5	$2 \times 0.5 = 1.0$	1

$$= (0.1101)_2 = 1 \times 2^{-1} + 1 \times 2^{-2} + 0 \times 2^{-3} + 1 \times 2^{-4}$$

$$(0.8125)_{10}$$

Multiply by 8	Integer
$0.8125 \times 8 = 6.5$	6
$0.5 \times 8 = 4.0$	4

$$(0.8125)_{10}$$

Multiply by 16	Integer
$0.8125 \times 16 = 13.0$	D

$$= (0.D)_{16}$$

$$= (0.64)_8$$

Octal to Binary & Binary to Octal

→ Every octal digit is represented by 3 bits.

$$0 - 000$$

$$1 - 001$$

$$2 - 010$$

$$3 - 011$$

$$4 - 100$$

$$5 - 101$$

$$6 - 110$$

$$7 - 111$$

$$(11011010110)_2$$

$$= (011 \ 011 \ 010 \ 110)_2$$

$$= (3 \ 3 \ 2 \ 6)_8$$

$$(124)_8$$

$$= (001 \ 010 \ 100)_2$$

Examples:-

→ Number Systems

$$724.5 = 7 \times 10^2 + 2 \times 10^1 + 4 \times 10^0 + 5 \times 10^{-1}$$

$$(312.4)_5 = 3 \times 5^2 + 1 \times 5^1 + 2 \times 5^0 + 4 \times 5^{-1} = 75 + 5 + 2 + 0.8 = (82.8)_{10}$$

→ Binary Numbers

$$(11010)_2 = 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 = 16 + 8 + 2 = (26)_{10}$$

Eg 2: Hexadecimal Addition

$$(59F)_{16} + (E46)_{16}$$

$$\begin{array}{r} 59F \\ + E46 \\ \hline \end{array}$$

Eg 4: Decimal Integers to Octal

$$153/8 = 19 + 1/8$$

$$19/8 = 2 + 3/8$$

$$2/8 = 0 + 2/8$$

$$(153)_{10} = (231)_8$$

least significant
Remainders = 1
3
2
most significant

Eg 5:- Decimal Integers to Binary

$41/2 = 20 + 1/2$	Remainders = 1	least
$20/2 = 10$	= 0	
$10/2 = 5$	= 0	
$5/2 = 2 + 1/2$	= 1	
$2/2 = 1$	= 0	
$1/2 = 0 + 1/2$	= 1	most

$$(41)_{10} = (101001)_2$$

Eg 6:- Decimal Fraction to Binary

$0.6875 \times 2 = 1.375$	Integer = 1	most
$0.375 \times 2 = 0.75$	= 0	
$0.75 \times 2 = 1.5$	= 1	
$0.5 \times 2 = 1$	= 1	least

$$(0.6875)_{10} = (0.1011)_2$$

Eg 7:- Decimal Fraction to Octal

$0.513 \times 8 = 4.104$	= 4
$0.104 \times 8 = 0.832$	= 0
$0.832 \times 8 = 6.656$	= 6
$0.656 \times 8 = 5.248$	= 5

$$(0.513)_{10} = (0.407)_8$$

Octal to Binary

Octal	2-bit
000 - 0	
001 - 1	
010 - 2	
011 - 3	
100 - 4	
101 - 5	
110 - 6	
111 - 7	

$$(318)_8 = (011001101)_2$$

$$(011011101)_2 = (335)_8$$

Hexa to Binary

0-F	4-bit	Unsigned	Signed
0	0000	0	0
1	0001	1	1
2	0010	2	2
3	0011	3	3
4	0100	4	4
5	0101	5	5
6	0110	6	6
7	0111	7	7
8	1000	8	-0
9	1001	9	-1
10	1010	10	-2
11	1011	11	-3
12	1100	12	-4
13	1101	13	-5
14	1110	14	-6
15	1111	15	-7

0100 1110
0110 0001

Aithmetic Operations

A	B	C	S
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

A	B	C	C	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

$$\begin{array}{r} 921 \\ 95 \\ \hline 1016 \end{array}$$

Addition:

$$\begin{array}{r} 1001 \\ 1011 \\ \hline 10100 \end{array}$$

Rules for binary subtraction:

$$\Rightarrow 0-0=0$$

$$1-0=1$$

$$1-1=0$$

$$0-1=-1 \times$$

$$x10x1011$$

$$-1101$$

$$1110$$

$$1101$$

$$1101$$

$$1110$$

$$1022$$

Multiplication:-

$$\begin{array}{r} 1101 \\ \times 101 \\ \hline 1101 \\ 0000 \\ 1101 \\ \hline 1000001 \end{array}$$

Division:-

$$101 \overline{)1101}$$

Signed Binary Numbers:-

- 1) Sign magnitude representation $\rightarrow$ MSB represent sign
- 2) 2's complement

$$\begin{array}{r} 0001 \\ +1 \\ \hline 1110 \\ -101 \\ \hline 1100 \end{array}$$

$$\begin{array}{l} 0 - +ve \\ 1 - -ve \end{array}$$

	Unsigned	Sign Mag
000	0	+0
001	1	+1
010	2	+2
011	3	+3
100	4	-0
101	5	-1
110	6	-2
111	7	-3

$$\begin{array}{r} 1 \\ -2 \\ -3 \\ \hline 0101 \\ 0110 \end{array}$$

MSB = Most Significant

2's Complement:-

+ve no. is started as it is

-ve no. is started after taking 2's complement

How to take 2's complement All 1's complement

11011

00100

-1's complement

MSB=0 +ve

MSB=1 -ve

+1

00101

⊗ Take 2's complement & replace it as 8-bit number

ⓐ +17

2	17
2	8 1
2	4 0
2	2 0
1	0

010001

For 8-bit, add 0 to front

00010001

11101110

+1

11101111

ⓑ -33

2	33
2	16 1
2	8 0
2	4 0
2	2 0
1	0

100001

00100001

11011110

-1's C

+1

11011111

-2's C

0 1 1 0 1 0 0 0
128 64 32 16 8 4 2 1
→ Consider only 1's & add them = (104)₁₀
Changed to decimal by using 2's C

Eg
11110101
00001010
+1
00001011
ⓐ ⓑ = 8+2+1 = (11)₁₀

Range of n-bit number

Unsigned → 0 → 2ⁿ-1

Sign-Magnitude → -(2ⁿ⁻¹-1) → 2ⁿ⁻¹-1

Two's Complement → 2ⁿ⁻¹ → 2ⁿ⁻¹-1

Basic Building Blocks of Digital Systems:-

1) AND Gate

2) OR Gate

3) NOT Gate

Representation of digital system

1) Logic diagram / Circuit diagram

2) Truth table

3) Algebraic Expression / Boolean Expression

2	185
2	92 1
2	46 0
2	23 0
2	11 1
2	5 1
2	2 1
1	0

10111001



Truth Tables

AND:- $\Rightarrow$

X	Y	Z = X · Y
0	0	0
0	1	0
1	0	0
1	1	1

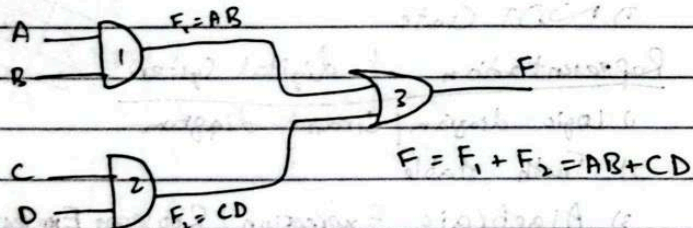
OR:- $\Rightarrow$

X	Y	Z = X + Y
0	0	0
0	1	1
1	0	1
1	1	1

NOT:- $\Rightarrow$

X	Z = $\bar{X}$
0	1
1	0

Algebraic Expressions



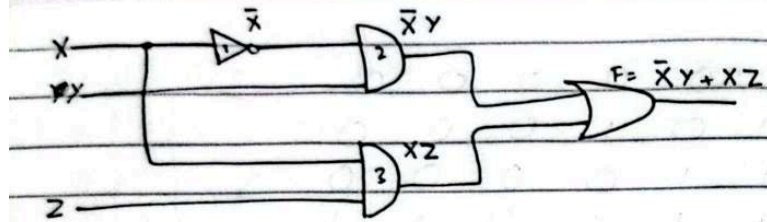
A	B	C	D	F ₁	F ₂	F
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	0	1	0	0	0	0
0	0	1	1	0	1	1

0	1	0	0	0	0	0
0	1	0	1	0	0	0
0	1	1	0	0	0	0
0	1	1	1	0	1	1
1	0	0	0	0	0	0
1	0	0	1	0	0	0
1	0	1	0	0	0	0
1	0	1	1	0	1	1
1	1	0	0	1	0	1
1	1	0	1	1	0	1
1	1	1	0	1	0	1
1	1	1	1	1	1	1

$$F(A, B, C, D) = \sum m(3, 7, 11, 12, 13, 14, 15)$$

Sum of minterms:-

Every row of fourth table corresponds to a unique minterm.



Find boolean expression

$$\Rightarrow F = \bar{X}Y + XZ$$

X	Y	Z	$\bar{X}$	$\bar{X}Y$	XZ	F
0	0	0	1	0	0	0
0	0	1	1	0	0	0
0	1	0	1	1	0	1
0	1	1	1	1	0	1
1	0	0	0	0	0	0
1	0	1	0	0	1	1
1	1	0	0	0	0	0
1	1	1	0	0	1	1

$$\frac{\bar{A}}{X} (\frac{\bar{B}+C}{Y}) \quad \bar{A} + BC$$

Axioms of Boolean Algebra:-

$$X+0=X$$

$$X+1=1$$

$$X \cdot 0 = 0$$

$$X \cdot 1 = X$$

$$X \cdot X = X$$

$$X \cdot \bar{X} = 0$$

$$X + X = X$$

$$X + \bar{X} = 1$$

$$\bar{\bar{X}} = X$$

$$X+Y=Y+X$$

$$XY=YX$$

$$X+(Y+Z)=(X+Y)+Z$$

$$(XY)Z=X(YZ)$$

$$X(Y+Z)=XY+XZ$$

$$X+YZ=(X+Y)(X+Z)$$

$$\overline{X+Y}=\bar{X}\bar{Y}$$

$$\overline{XY}=\bar{X}+\bar{Y}$$

Commutative

Associative

Distributive

De-Morgan's law

X	Y	Z	YZ	X+YZ	X+Y	X+Z	(X+Y)(X+Z)
0	0	0	0	0	0	0	0
0	0	1	0	0	0	1	0
0	1	0	0	0	1	0	0
0	1	1	1	1	1	1	1
1	0	0	0	1	1	1	1
1	0	1	0	1	1	1	1
1	1	0	0	1	1	1	1
1	1	1	1	1	1	1	1

20/8/20
20/8/20
20/8/20

→ Minimize the given expression using boolean algebra.

→ Two different logic circuits and different boolean expressions can correspond to same truth table.

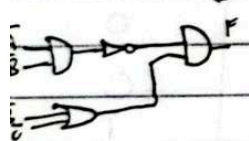
→ Simplified version of boolean expression by algebraic manipulation is not unique.

→ Simplified form may not be the most optimal one.

Q) Minimize the given expression using boolean algebra.

$$F = (\overline{A}B) \cdot (\overline{B} + \overline{C})$$

~~$$F = (\overline{A}B) \cdot (\overline{B} + \overline{C})$$~~

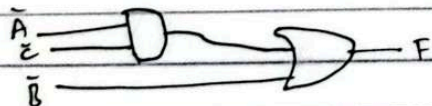


$$\overline{A} \cdot \overline{B} \cdot (\overline{B} + \overline{C})$$

$$B = \overline{A} \cdot \overline{C}$$

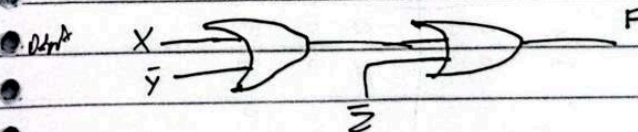
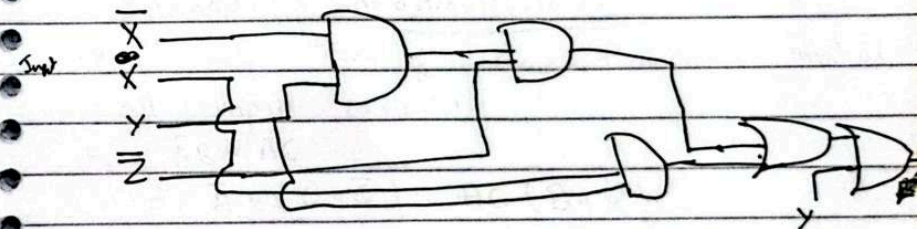
~~$$B = \overline{A} \cdot \overline{C}$$~~

$$\begin{aligned} F &= (\overline{A} + \overline{B}) \cdot (\overline{B} + \overline{C}) \\ &= \overline{A} \cdot \overline{B} + \overline{A} \cdot \overline{C} + \overline{B} \cdot \overline{B} + \overline{B} \cdot \overline{C} \\ &= \overline{A} \cdot \overline{B} + \overline{A} \cdot \overline{C} + \overline{B} + \overline{B} \cdot \overline{C} \\ &= \overline{A} \cdot \overline{B} + \overline{A} \cdot \overline{C} + \overline{B} (1 + \overline{C}) \\ &= \overline{A} \cdot \overline{B} + \overline{A} \cdot \overline{C} + \overline{B} \\ &= \overline{B} + \overline{A} \cdot \overline{B} + \overline{A} \cdot \overline{C} \\ &= \overline{B} (1 + \overline{A}) + \overline{A} \cdot \overline{C} = \overline{B} + \overline{A} \cdot \overline{C} \end{aligned}$$



Q) Assume that inputs are available in true and complemented form.

$$\begin{aligned} F &= \overline{Y} + \overline{X} Y \overline{Z} + X Y \overline{Z} \\ &= \overline{Y} + Y (X + \overline{X} \overline{Z}) \\ &= \overline{Y} + Y (X + \overline{X}) (X + \overline{Z}) \\ &= \overline{Y} + Y (X + \overline{Z}) \\ &= \overline{Y} + Y (X + \overline{Z}) \\ &= \overline{Y} + Y X + Y \overline{Z} \\ &= (\overline{Y} + Y) (\overline{Y} + X) + Y \overline{Z} \\ &= X + \overline{Y} + Y \overline{Z} \\ &= X + (\overline{Y} + Y) (\overline{Y} + \overline{Z}) \\ &= X + \overline{Y} + \overline{Z} \end{aligned}$$



$$\begin{aligned}
 Q) F &= \bar{W} \times (\bar{Z} + \bar{Y}Z) + X(W + \bar{W}YZ) \\
 &\therefore A+BC = (A+B)(A+C) \\
 &= \bar{W} \times (\bar{Z} + \bar{Y})(\bar{Z} + Z) + X(W + \bar{W})(W + YZ) \\
 &= \bar{W} \times (\bar{Z} + \bar{Y}) + XW + XYZ \\
 &= \bar{W} \times \bar{Z} + \bar{W} \times \bar{Y} + XW + XYZ \\
 &= X[\bar{W}\bar{Z} + \bar{W}\bar{Y} + W + YZ] \\
 &= X[\bar{W}\bar{Z} + (W + \bar{W})(W + \bar{Y}) + YZ] \\
 &= X[\bar{W}\bar{Z} + W + \bar{Y} + YZ] \\
 &= X[(W + \bar{W})(W + \bar{Z}) + (\bar{Y} + Y)(\bar{Y} + Z)] \\
 &= X[W + \bar{Z} + \bar{Y} + Z] \\
 &= X[W + \bar{Y} + 1] \\
 &= X[W + 1] \\
 &= X
 \end{aligned}$$

Q) Find the complement of given function

$$\begin{aligned}
 \bar{F} &= (A + \bar{B} + C)(\bar{A}\bar{B} + \bar{C})(A + \bar{B}\bar{C}) \\
 &= (A + \bar{B} + C) + (\bar{A}\bar{B} + \bar{C}) + (A + \bar{B}\bar{C}) \\
 &= (\bar{A} \cdot \bar{B} \cdot \bar{C}) + (\bar{A}\bar{B} \cdot \bar{C}) + (\bar{A} \cdot \bar{B} \cdot \bar{C}) \\
 &= \bar{A}\bar{B}\bar{C} + (\bar{A} + \bar{B}) \cdot \bar{C} + \bar{A}(\bar{B} + \bar{C}) \\
 &= \bar{A}\bar{B}\bar{C} + \bar{A}\bar{C} + \bar{B}\bar{C} + \bar{A}\bar{B} + \bar{A}\bar{C}
 \end{aligned}$$

$$\begin{aligned}
 \bar{F} &= \bar{A}\bar{B}\bar{C} + \bar{A}\bar{C} + \bar{B}\bar{C} + \bar{A}\bar{B} + \bar{A}\bar{C} \\
 &= (\bar{A}\bar{B}\bar{C}) + (\bar{A}\bar{C}) + (\bar{B}\bar{C}) + (\bar{A}\bar{B}) + (\bar{A}\bar{C}) \\
 &= (\bar{A} + \bar{B} + \bar{C})(\bar{A} + \bar{C})(\bar{B} + \bar{C})(\bar{A} + \bar{B})(\bar{A} + \bar{C}) \\
 &= (A + \bar{B} + C)(\bar{A} + \bar{C})(\bar{B} + \bar{C})(A + \bar{B})(A + \bar{C}) \\
 &= (A + \bar{B} + C)(\bar{C} + \bar{A}\bar{B})(A + \bar{B}\bar{C})
 \end{aligned}$$

Standard (Canonical) Form:-

→ if each term of the function involves all input variables

$$\begin{aligned}
 F_2 &= \bar{A}\bar{B} + AC \\
 &= \bar{A}\bar{B}(C + \bar{C}) + AC(B + \bar{B}) \\
 &= \bar{A}\bar{B}C + \bar{A}\bar{B}\bar{C} + ABC + A\bar{B}\bar{C}
 \end{aligned}$$

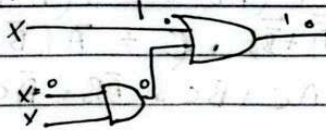
A	B	C	F ₂		
0	0	0	0	m ₀	
0	0	1	0	m ₁	
0	1	0	1	m ₂	
0	1	1	1	m ₃	
1	0	0	0	m ₄	
1	0	1	1	m ₅	
1	1	0	0	m ₆	
1	1	1	1	m ₇	

$F(A, B, C) = \sum m(2, 3, 5, 7)$

Absorption Law:-

$$X + XY = X$$

$$(X+Y)(X+Z)$$



⇒ Duality holds in Boolean Algebra

Convert $Y \rightarrow 0$

$$0 \rightarrow 1$$

$$+ \rightarrow \cdot$$

$$\cdot \rightarrow +$$

$$X + 0 = X$$

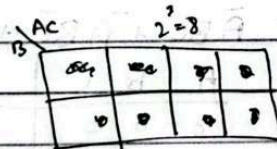
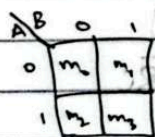
$$X \cdot 1 = X$$

K-Map (Karnaugh Map):-

⇒ graphical method to reduce the boolean expression.

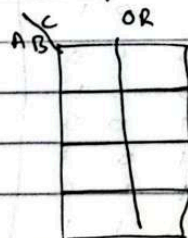
2, 3, 4 inputs

Prime algo / Mc Cluskey Methode



A B

0	0	0
0	1	1
1	0	1
1	1	1



* Make group of neighbouring one's (1's)

* Write algebraic expression for each group

* Each expression will be a product term.

* Add all the expression to get final result

* While writing expression, the literal it will not be considered if both values (1 & 0) are involved

⇒ Rules for making groups:-

1) Group size should be power of 2.

2) Groups can be made with neighbouring 1's only.

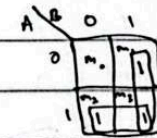
3) Groups size should be as large as possible.

4) We can't go diagonal.

5) There must be atleast unique element in each group.

6) Groups can overlap.

7) Total no. of groups should be as small as possible

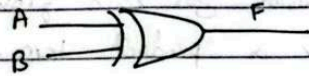


$$G_1(1,3) = B$$

$$G_2(2,3) = A$$

$$F = A + B$$

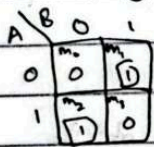
Two Input XOR (Exclusive OR):-



A	B	F
0	0	0
0	1	1
1	0	1
1	1	0

$$F = \bar{A}B + A\bar{B}$$

$$= A \oplus B$$



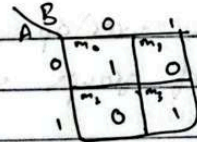
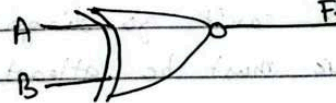
$$G_1(1) = \bar{A}B$$

$$G_2(2) = A\bar{B}$$

$$F = \bar{A}B + A\bar{B}$$

XNOR:-

A	B	F
0	0	1
0	1	0
1	0	0
1	1	1



$$G_1(0) = \bar{A}\bar{B}$$

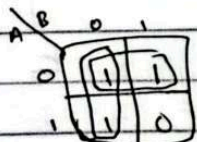
$$G_2(3) = AB$$

$$F = \bar{A}\bar{B} + AB$$

$$F = \bar{A}\bar{B} + AB$$

$$= A \odot B$$

Two Input NAND:-

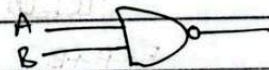


$$G_1(0,1) = \bar{A}$$

$$G_2(0,2) = \bar{B}$$

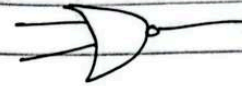
$$F = \bar{A} + \bar{B}$$

$$F = \overline{AB}$$

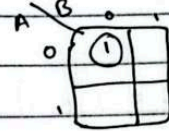


A	B	F
0	0	1
0	1	1
1	0	1
1	1	0

Two Input NOR:-



A	B	F
0	0	1
0	1	0
1	0	0
1	1	0

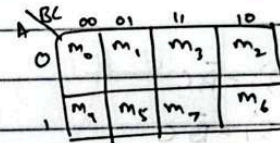


$$G_1(0) = \bar{A}\bar{B}$$

$$F = \bar{A}\bar{B}$$

$$XF = \overline{A+B}$$

Three Input K-Map:-



Gray-Code

⇒ B/w two neighbouring sequences, there is change of 1-bit only.

A	B	C	
0	0	0	m_0
0	0	1	m_1
0	1	0	m_2
0	1	1	m_3
1	0	0	m_4
1	0	1	m_5
1	1	0	m_6
1	1	1	m_7

Q) $F = \sum m(1, 3, 5, 7)$

BC	00	01	11	10
A				
0		1	1	
1		1	1	

$\therefore$ Group size as large as possible

$G_1(1, 3, 5, 7) = C$

$F = C$

Q) $F = F(A, B, C) = \sum m(0, 1, 2, 3, 4, 5)$

BC	00	01	11	10
A				
0	1	1	1	1
1	1	1		

$G_1(0, 1, 2, 3) = \bar{A}$

$G_2(0, 1, 4, 5) = \bar{B}$

$F = \bar{A} + \bar{B}$

Q) $F(A, B, C) = \sum m(1, 3, 4, 5, 6)$

BC	00	01	11	10
A				
0			1	1
1	1	1	1	1

$G_1(1, 3) = \bar{A}C$

$G_2(1, 5) = \bar{B}C$

$G_3(4, 6) = A\bar{C}$

$F = \bar{A}C + \bar{B}C + A\bar{C}$

Q) $F(A, B, C) = \sum m(0, 1, 2, 4, 5, 6)$

BC	00	01	11	10
A				
0	1	1		1
1	1	1		1

$G_1(0, 1, 4, 5) = \bar{B}$

$G_2(0, 2, 4, 6) = \bar{C}$

$F = \bar{B} + \bar{C}$

4- Input K-Map:-

AB\CD	00	01	11	10
00	m_0	m_1	m_3	m_2
01	m_4	m_5	m_7	m_6
11	m_8	m_9	m_{11}	m_{10}
10	m_{12}	m_{13}	m_{15}	m_{14}

$11 = 0$

$00 = \bar{C}$

Q) $F(A, B, C) = \sum m(0, 1, 2, 4, 5, 6, 8, 9, 10, 12, 13)$

AB\CD	00	01	11	10
00	1	1		1
01	1	1		1
11	1	1		
10	1	1		1

$G_1(0, 1, 4, 5, 8, 9, 12, 13) = \bar{C}$

$G_2(0, 2, 4, 6) = \bar{A}\bar{D}$

$G_3(0, 2, 8, 10) = \bar{B}\bar{D}$

$F = \bar{C} + \bar{A}\bar{D} + \bar{B}\bar{D}$

Implicant:-

Any group of 1's.

Prime Implicants:-

largest possible group of 1's.

Essential Prime Implicants:-

A prime implicant is called essential prime implicant if there is atleast a single '1' that cannot be combined in any other group.

AB \ CD	00	01	11	10
00	1	1	1	0
01	1	0	0	1
11	1	1	0	0
10	1	0	1	1

CD	00	01	11	10
00	1	1	1	0
01	1	0	0	1
11	1	1	0	0
10	1	0	1	1

AB \ CD	00	01	11	10
00	1	1	1	0
01	1	0	0	1
11	1	1	0	0
10	1	0	1	1

$$G_1(0, 4, 8, 12) =$$

CD	00	01	11	10
00	1	1	1	0
01	1	0	0	1
11	1	1	0	0
10	1	0	1	1

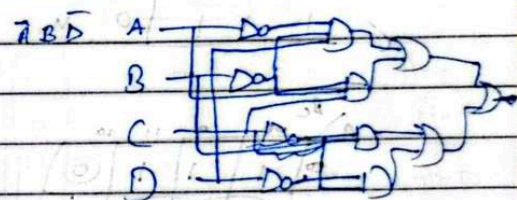
$$G_2(1, 3) =$$

AB \ CD	00	01	11	10
00	1	1	1	0
01	1	0	0	1
11	1	1	0	0
10	1	0	1	1

$$G_3(10, 11) =$$

$$G_4(6, 4) =$$

$$= \bar{A}\bar{B}D + \bar{A}B\bar{C} + \bar{C}\bar{A}C\bar{D}$$



$$G_5(12, 13) =$$

$$= \bar{A}B\bar{C}$$

$$F = \bar{A}\bar{B}D + \bar{A}B\bar{C} + \bar{A}B\bar{C} + \bar{A}B\bar{C} + \bar{C}\bar{D}$$

$$= \bar{A}\bar{B}D + \bar{A}B\bar{C} + \bar{C}\bar{D}$$

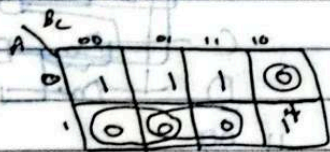
$$= \bar{A}\bar{B}D + \bar{A}B\bar{C} + \bar{C}(\bar{B} + \bar{D})$$

Concept of Maxterms:-

A	B	C	F
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

$$F(A,B,C) = \sum m(0,1,3,5) \quad \text{SOP}$$

$$F(A,B,C) = \prod M(2,4,5,7) \quad \text{POS}$$



$$(A + \bar{B} + C)(\bar{A} + B)$$

$$F = (A + \bar{B} + C)(\bar{A} + B)(\bar{A} + \bar{C})$$

$$F = (\bar{A} + B)(A + C)$$

$$= (\bar{A} + B + C\bar{C})(A + C + B\bar{B})$$

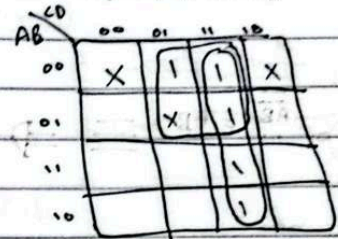
$$= (\bar{A} + B + C)(\bar{A} + B + \bar{C})(A + B + C)(A + \bar{B} + C)$$

$$= \prod M(0,2,4,5)$$

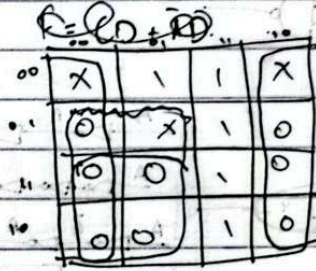
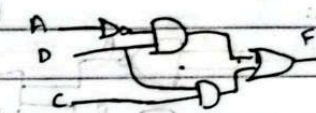
Don't Care Conditions:-

$$F(A,B,C,D) = \sum m(1,3,7,11,15)$$

$$\phi(A,B,C,D) = \sum m(0,2,5)$$



$$F = CD + \bar{A}D$$

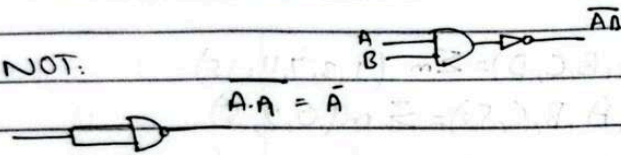


$$F = (\bar{A} + C)D$$

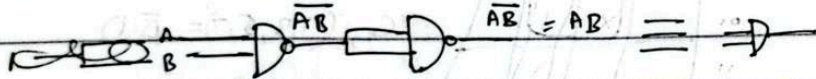


NAND-NAND Implementation:-

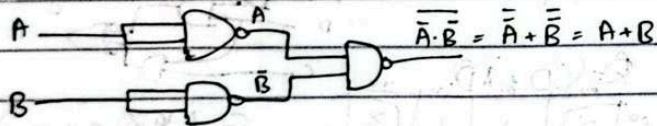
NOT:



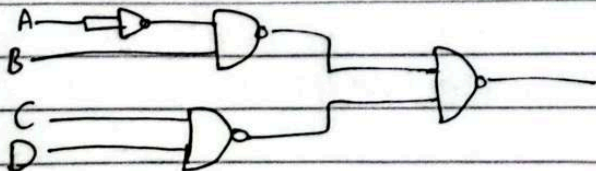
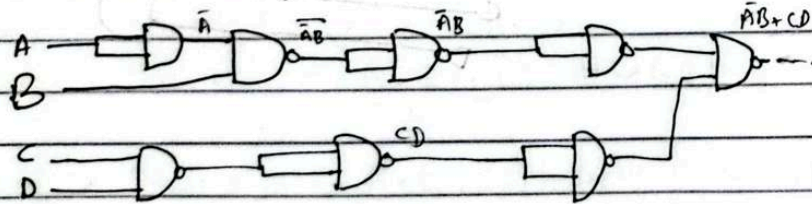
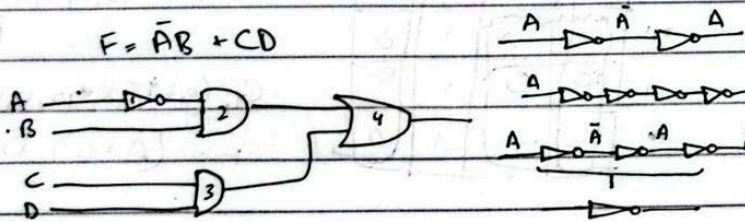
AND:



OR:

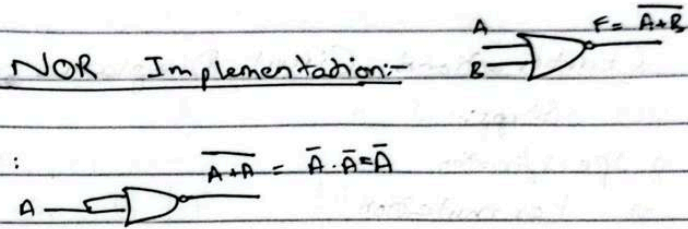


$$F = \bar{A}B + CD$$

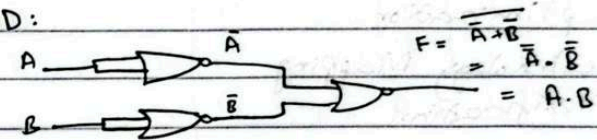


NOR-NOR Implementation:-

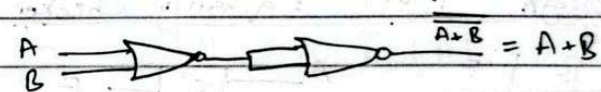
NOT:



AND:

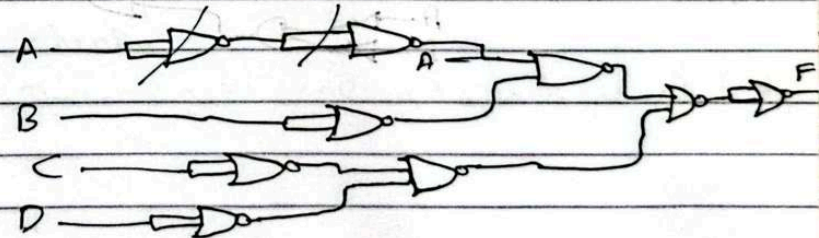
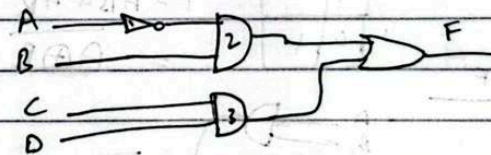


OR:



~~NOT~~

$$F = \bar{A}B + CD$$



Combinational Circuit Diagram

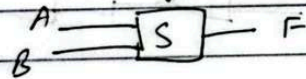
Steps:-

- 1) Specification
- 2) Formulation
- 3) Optimization
- 4) Technology Mapping
- 5) Verification

Design 1-bit Equality Checker

Equal = 1
Not Equal = 0

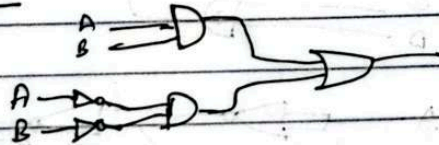
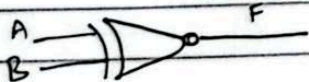
A	B	F
0	0	1
0	1	0
1	0	0
1	1	1



A	B
0	1
1	0

$$F = \bar{A}\bar{B} + AB$$

$$= A \oplus B$$



Q) Design a digital system that takes a 3-bit no. as input and checks whether the number of 1's in the pattern are odd or even.

A ₂	A ₁	A ₀	F
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

A ₂ A ₁ A ₀	00	01	10	11
0	0	1	0	1
1	1	0	1	0

$$F = \bar{A}_2\bar{A}_1\bar{A}_0 + \bar{A}_2\bar{A}_1A_0 + \bar{A}_2A_1\bar{A}_0 + \bar{A}_2A_1A_0 + A_2\bar{A}_1\bar{A}_0 + A_2\bar{A}_1A_0 + A_2A_1\bar{A}_0 + A_2A_1A_0$$

$$= \bar{A}_2(\bar{A}_1\bar{A}_0 + \bar{A}_1A_0 + A_1\bar{A}_0 + A_1A_0)$$

$$= \bar{A}_2(\bar{A}_1(\bar{A}_0 + A_0) + A_1(\bar{A}_0 + A_0))$$

$$= \bar{A}_2(\bar{A}_1 \cdot 1 + A_1 \cdot 1)$$

$$= \bar{A}_2(\bar{A}_1 + A_1)$$

$$= \bar{A}_2(1)$$

$$= \bar{A}_2$$

$$\text{Let } A_2 = x, A_1 \oplus A_0 = y$$

$$= A_2 \oplus (A_1 \oplus A_0)$$

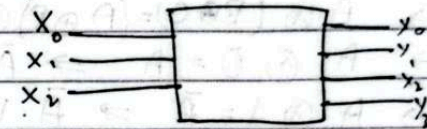
⇒ Two input XNOR acts as 1-bit equality checker

⇒ Three input XOR gates acts as odd function.

Q) Design a system that takes a 3-bit number as input. Design a digital system that implements this scenario:-

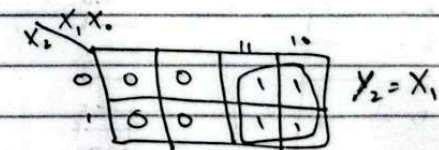
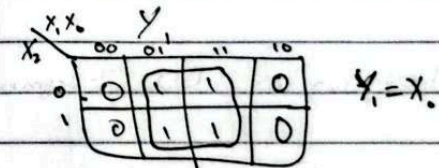
A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

Q) Design a doubler that takes a 3-bit number as input.



x_2	x_1	x_0	y_2	y_1	y_0
0	0	0	0	0	0
0	0	1	0	0	1
0	1	0	0	1	0
0	1	1	0	1	1
1	0	0	1	0	0
1	0	1	1	0	1
1	1	0	1	1	0
1	1	1	1	1	1

$$y_0 = 0$$



$$y_2 = x_2$$

Properties of XOR:-

- $A \oplus B = B \oplus A$
- $A \oplus (B \oplus C) = (A \oplus B) \oplus C$
- $A \oplus 0 = A \Rightarrow \bar{A} \cdot 0 + A \cdot \bar{0} = A$ Buffer
- $A \oplus 1 = \bar{A} \Rightarrow \bar{A} \cdot 1 + A \cdot \bar{1} = \bar{A}$ Inverter
- $A \oplus A = 0$
- $A \oplus \bar{A} = 1$



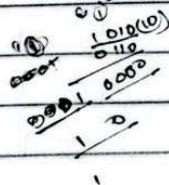
Binary Coded Decimal:-

(2814)₁₀

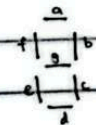
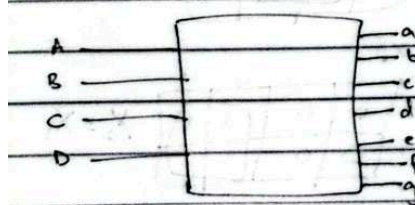
= (0010 1000 0001 0100)_{BCD}

Add BCD 6 (0110) to any number above 9 to convert to BCD

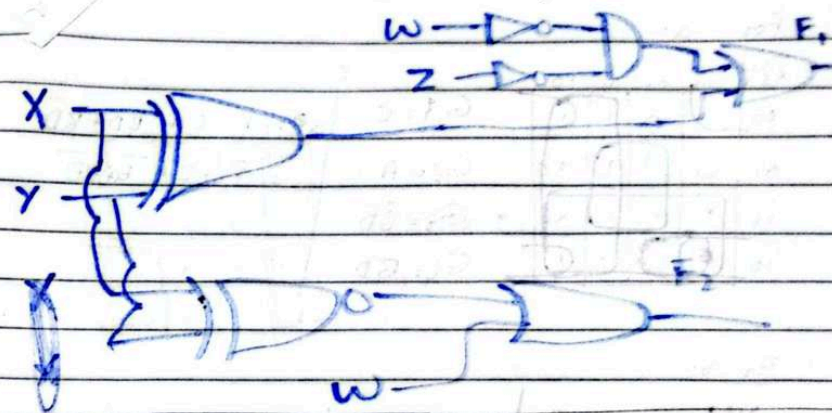
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001



Designing of BCD 7-segment display decoder



9



	A	B	C	D	a	b	c	d	e	f	g	
0	0	0	0	0	1	1	1	1	1	1	0	a
1	0	0	0	1	0	1	1	0	0	0	0	f g c
2	0	0	1	0	1	1	0	1	1	0	1	e c
3	0	0	1	1	1	1	1	1	0	0	1	d
4	0	1	0	0	0	1	1	0	0	1	1	
5	0	1	0	1	1	0	1	1	0	1	1	
6	0	1	1	0	1	0	1	1	1	1	1	
7	0	1	1	1	1	1	1	0	0	0	0	
8	1	0	0	0	1	1	1	1	1	1	1	
9	1	0	0	1	1	1	1	1	0	1	1	
	1	0	1	0	x	x	x	x	x	x	x	Don't care table
	1	0	1	1	x	x	x	x	x	x	x	
	1	1	0	0	x	x	x	x	x	x	x	
	1	1	0	1	x	x	x	x	x	x	x	
	1	1	1	0	x	x	x	x	x	x	x	
	1	1	1	1	x	x	x	x	x	x	x	

XOR: $F = \bar{A}B + AB$
XNOR: $F = \bar{A}\bar{B} + AB$

For a:-

AB \ CD	00	01	11	10	
00	1	0	1	1	$G_1 = C$
01	0	1	1	1	$G_2 = A$
11	x	x	x	x	$G_3 = \bar{B}\bar{D}$
10	1	1	x	x	$G_4 = BD$

$$a = A + C + \bar{B}\bar{D} + BD$$

$$= A + C + \overline{B\bar{D}} + BD$$

For b:-

AB \ CD	00	01	11	10	
00	1	1	1	1	$G_1 = \bar{B}$
01	1	0	1	0	$G_2 = \bar{C}\bar{D}$
11	x	x	x	x	$G_3 = CD$
10	1	1	x	x	

$$b = \bar{B} + \bar{C}\bar{D} + CD$$

$$= \bar{B} + \overline{C\bar{D}}$$

For c:-

AB \ CD	00	01	11	10	
00	1	1	1	0	$G_1 = \bar{C}$
01	1	1	1	1	$G_2 = D$
11	x	x	x	x	$G_3 = B$
10	1	1	x	x	

$$c = B + \bar{C} + D$$

For d:-

AB \ CD	00	01	11	10	
00	1	0	1	1	$G_1 = \bar{B}\bar{D}$
01	0	1	0	1	$G_2 = B\bar{C}D$
11	x	x	x	x	$G_3 = A$
10	1	1	x	x	$G_4 = \bar{B}C$
					$G_5 = \bar{C}D$

$$d = A + \bar{B}C + B\bar{C}D + \bar{B}\bar{D} + \bar{C}D$$

For e:-

AB \ CD	00	01	11	10	
00	1	0	0	1	$G_1 = \bar{B}\bar{D}$
01	0	0	0	1	$G_2 = \bar{A}\bar{D}$
11	x	x	x	x	$G_3 = \bar{C}\bar{D}$
10	1	0	x	x	

$$e = \bar{B}\bar{D} + \bar{C}\bar{D}$$

$$= (\bar{B} + \bar{C})\bar{D}$$

For f:-

AB \ CD	00	01	11	10	
00	1	0	0	0	$G_1 = \bar{C}\bar{D}$
01	1	1	0	1	$G_2 = B\bar{C}$
11	x	x	x	x	$G_3 = B\bar{D}$
10	1	1	x	x	$G_4 = A$

$$f = A + B\bar{C} + B\bar{D} + \bar{C}\bar{D}$$

$$= A + B(\bar{C} + \bar{D}) + \bar{C}\bar{D}$$

For g:-

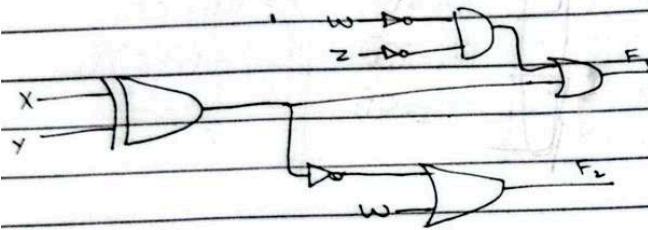
AB \ CD	00	01	11	10	
00	0	0	1	1	$G_1 = A$
01	1	1	0	1	$G_2 = \bar{B}C$
11	x	x	x	x	$G_3 = \bar{B}\bar{C}$
10	1	1	x	x	$G_4 = \bar{C}\bar{D}$

$$g = A + \bar{B}\bar{C} + \bar{B}C + \bar{C}\bar{D}$$

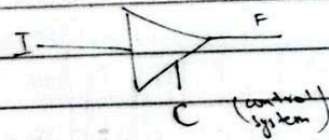
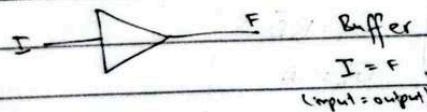
$$= A + B\bar{C} + \bar{C}\bar{D}$$

$$F_1 = \bar{x}y + x\bar{y} + \bar{w}z = x \oplus y + \bar{w}z$$

$$F_2 = xy + \bar{x}\bar{y} + w = \bar{x} \oplus y + w$$

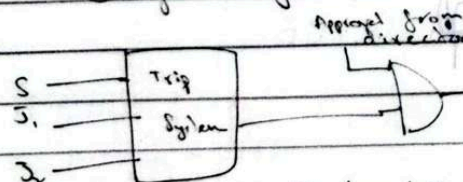


Concept of Tri-State Buffer:-



C	I	F
0	X	High impedance
1	0	0
1	1	1

Enable Signal:-

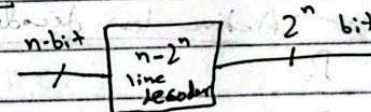


S	J ₁	J ₂	C	F
0	0	0	0	
0	0	1	0	
0	1	0	0	
0	1	1	0	
1	0	0	1	
1	0	1	1	
1	1	0	1	
1	1	1	1	

Design of Combinational circuits using Functional Blocks:-

- 1) Decoders -
- 2) Encoders -
- 3) Multiplexer -
- 4) De Multiplexer -
- 5) Comparator -
- 6) Adder -
- 7) Subtractor -
- 8) Multiplier -

1) Decoder



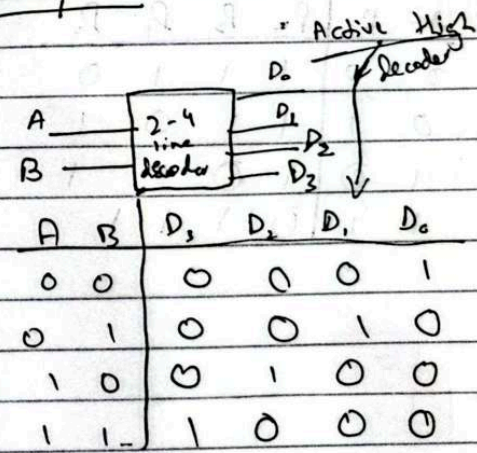
→ 1-2 line decoder
1x2

→ 2-4
2x4

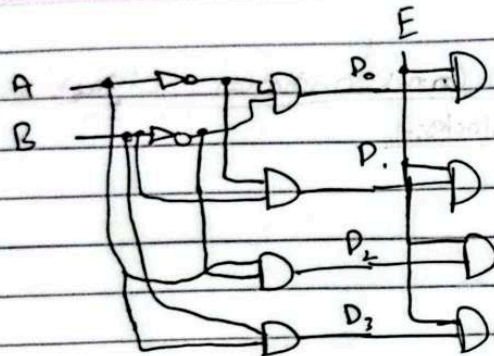
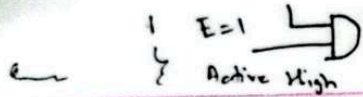
→ 3x8

→ 4x16

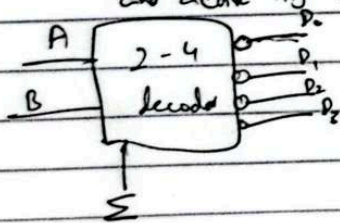
→ 5-32



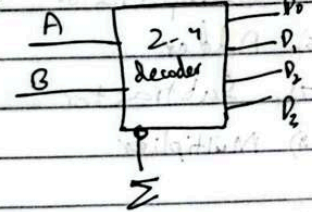
A	B	D ₃	D ₂	D ₁	D ₀
0	0	0	0	0	1
0	1	0	0	1	0
1	0	0	1	0	0
1	1	1	0	0	0



Active low output and active high enable



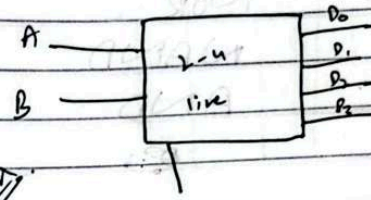
Active high output and active low enable



Active low decoder

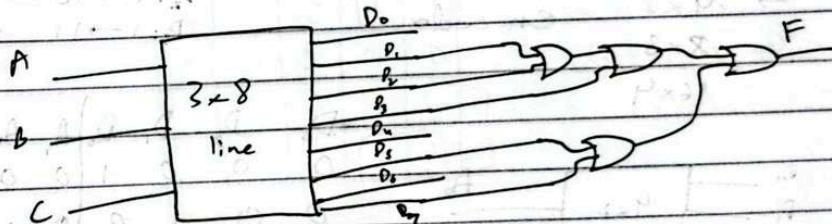
A	B	P ₃	P ₂	P ₁	P ₀
0	0	1	1	1	0
0	1	1	1	0	1
1	0	1	0	1	1
1	1	0	1	1	1

Q) $F(A, B) = \sum m(0, 2) = m_0 + m_2$

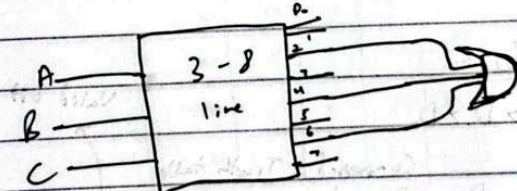


A	B	F
0	0	1
0	1	0
1	0	1
1	1	0

Q) $F(A, B, C) = \sum m(1, 2, 3, 5, 7)$



Q) $F(A, B, C) = \prod M(2, 4, 6)$



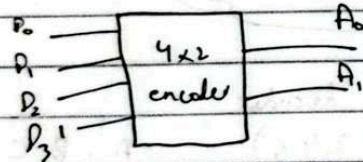
A	B	C	F
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

min-3
 minterm Active high decoder OR
 Maxterm " high " NOR
 minterm " low " NAND
 Maxterm " low " AND

2) Encoders

- opposite of decoder
- 2^n - inputs, n - outputs
- 4×2 encoder
- 8×3
- 16×4

1/1 0/1
 $D_3: 1 \rightarrow 00$
 $D_2: 1 \rightarrow 01$
 $D_1: 1 \rightarrow 10$
 $D_0: 1 \rightarrow 11$



D_3	D_2	D_1	D_0	A_1	A_0
0	0	0	0	0	0
0	0	1	0	0	1
0	1	0	0	1	0
1	0	0	0	1	1

Priority Encoder:-

$D_3 > D_2 > D_1 > D_0$

Compressed Truth table

D_3	D_2	D_1	D_0	A_1	A_0	V
0	0	0	1	0	0	1
0	0	1	0	0	1	1
0	1	0	0	1	0	1
1	0	0	0	1	1	1
0	0	0	0	X	X	0

For A_1 :-

D_3	D_2	D_1	D_0
X	0	0	0
1	1	1	1
1	1	1	1
1	1	1	1

$$G_1(8) = D_3$$

$$G_2(8) = D_3$$

$$F = D_3 + D_3$$

For A_0 :-

D_3	D_2	D_1	D_0
X	0	1	1
0	0	0	0
1	1	1	1
1	1	1	1

$$G_1(8) = D_3$$

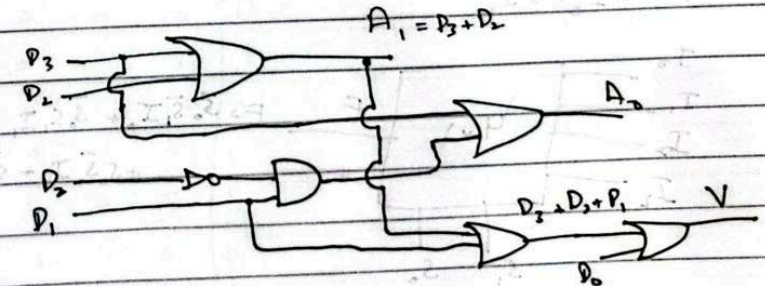
$$G_2(4) = \overline{D_2} + D_1$$

$$F = D_3 + D_1 \overline{D_2}$$

For V :-

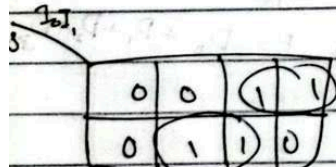
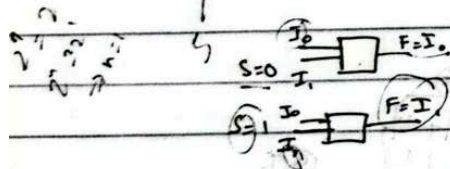
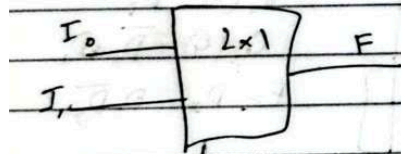
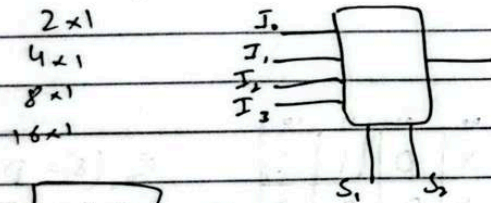
D_3	D_2	D_1	D_0
0	1	1	1
1	1	1	1
1	1	1	1
1	1	1	1

$$F = D_0 + D_1 + D_2 + D_3$$

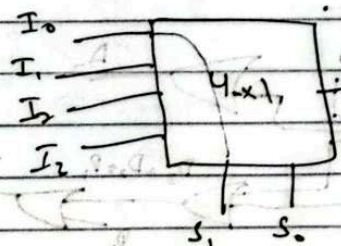


3) Multiplexer (MUX)

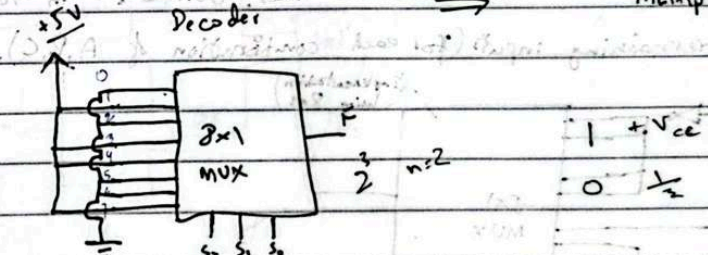
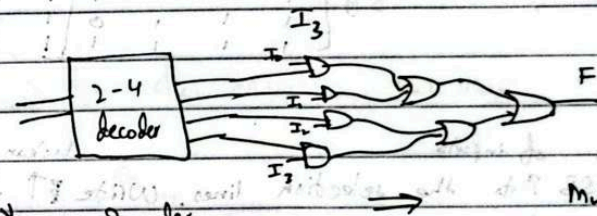
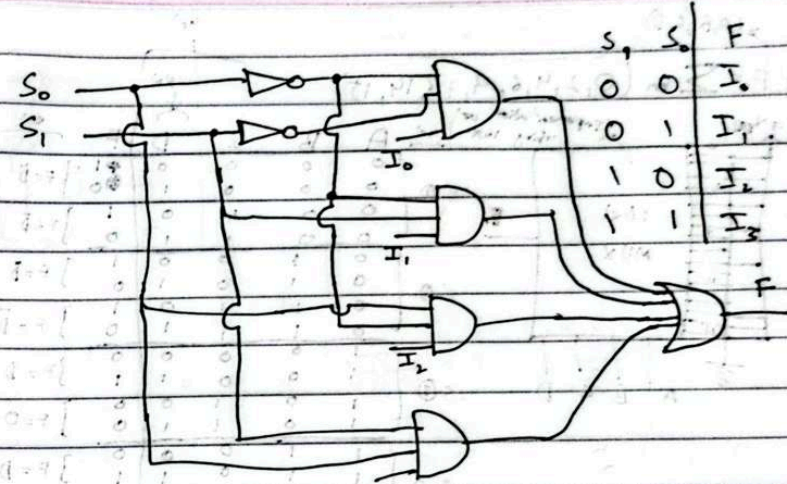
- Multiple inputs — single output
- Input lines — 2^n
- Selection line — n
- Output — 1



$$F = S_1 I_0 + S_1 I_1$$



$$F = S_0 \bar{S}_1 I_0 + S_0 S_1 I_1 + S_1 \bar{S}_0 I_2 + S_1 S_0 I_3$$

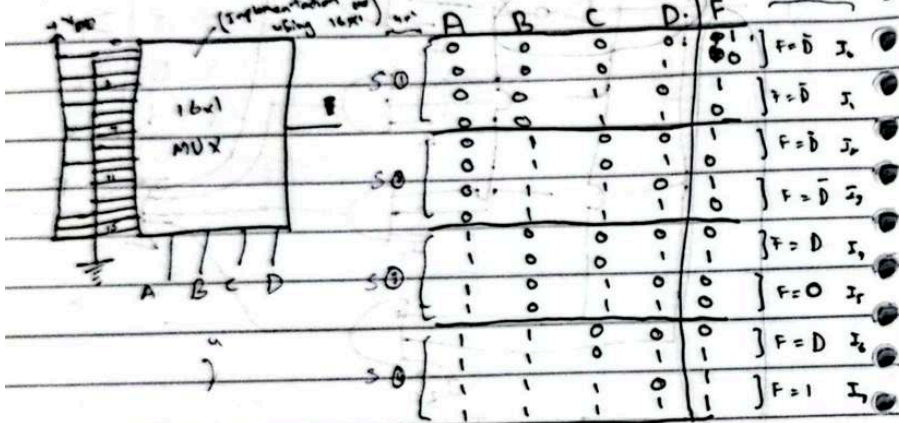


S_2	S_1	S_0	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

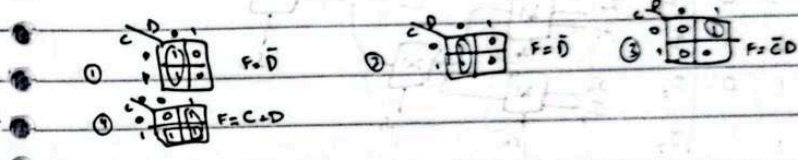
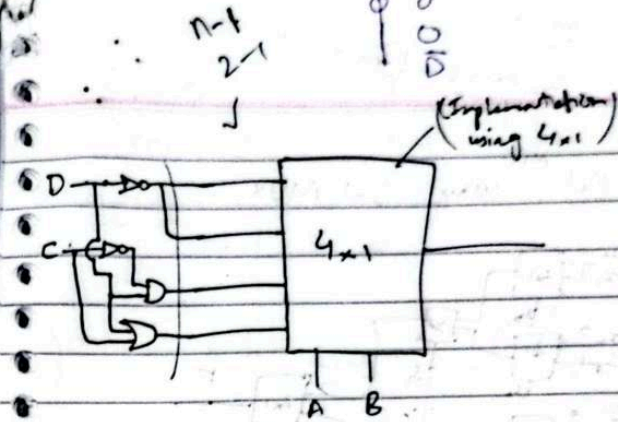
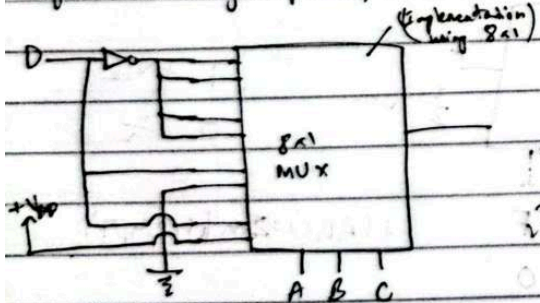
$$F(A, B, C) = \sum m(1, 3, 4, 7)$$

$> (A, B, C, D)$

Q) $F = \sum m(0, 2, 4, 6, 9, 13, 14, 15)$

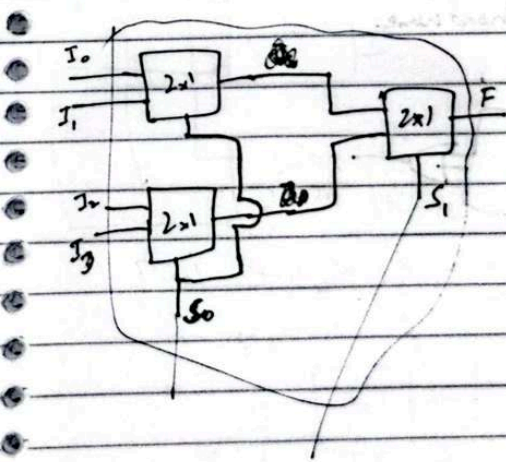


(ii) of inputs
Apply MSB to the selection lines. Write F in terms of remaining inputs (for each combination of A, B, C).



MUX Trees:-

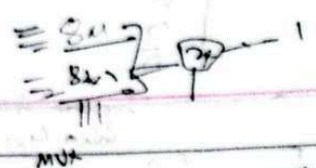
a) Constructed 4x1 MUX using 2x1 MUX



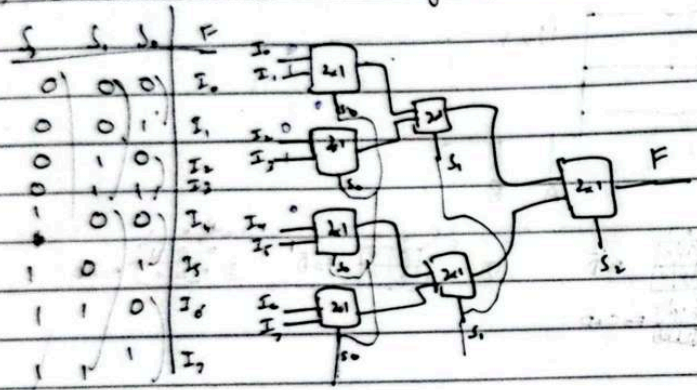
S_1	S_0	F
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

25

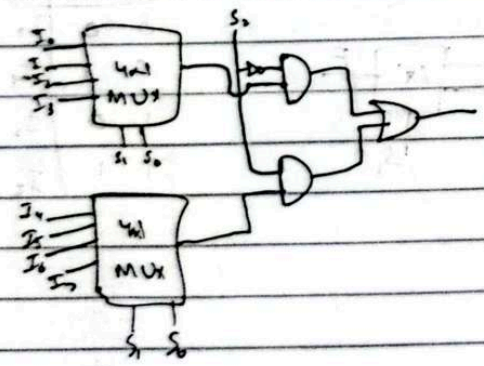
10x1



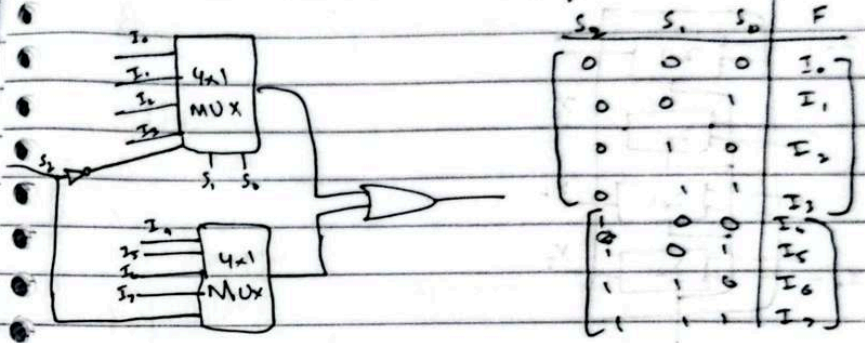
Q) Construct 8x1 using 2x1 MUX



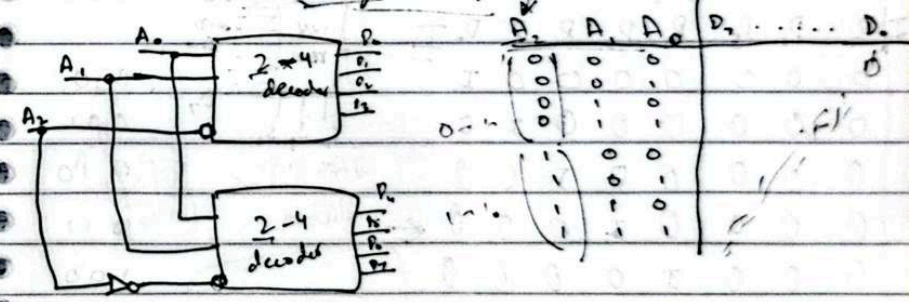
2) Construct 8x1 MUX using 4x1 MUX (without enable) and min extra hardware.



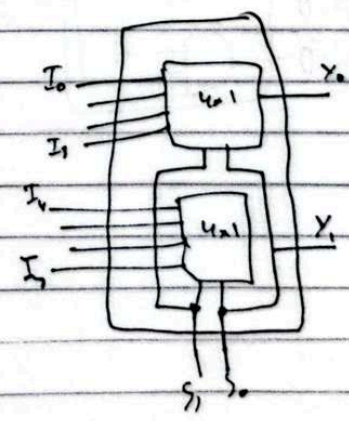
Q) Construct 8x1 MUX using 4x1 MUX with enable



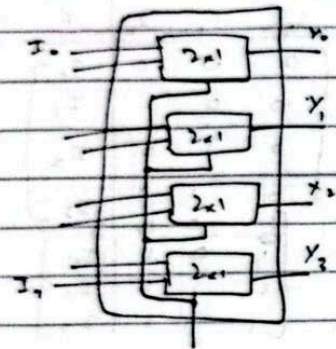
Q) Construct 3-8 line decoder using 2-4 line decoder with active high output and active low enable.



Dual 4x1 MUX:-

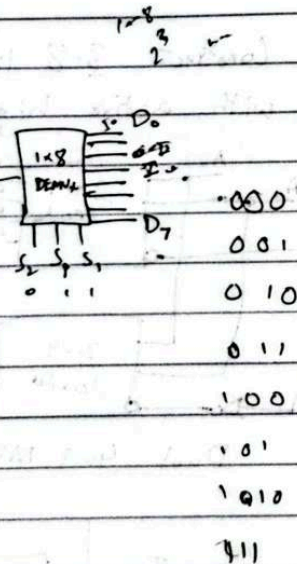


Ques 2x1 MUX:-



4) De Multiplexer:-

1x2		1x4		1x8		1x16	
D ₇	D ₆	D ₅	D ₄	D ₃	D ₂	D ₁	D ₀
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	1
0	0	0	0	0	0	1	0
0	0	0	0	0	1	0	0
0	0	0	0	1	0	0	0
0	0	0	1	0	0	0	0
0	0	1	0	0	0	0	0
0	1	0	0	0	0	0	0
1	0	0	0	0	0	0	0



Arithmetic Operations:-

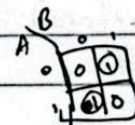
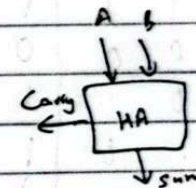
- 6) Adders
- 7) Subtractors
- 8) Multiplier

6) Adders:-

Half-Adder:-

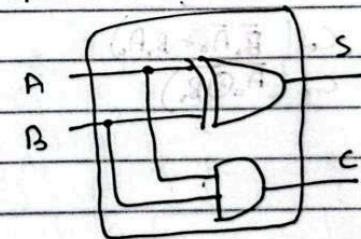
A	B	C	S
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

Adds two 1-bit numbers



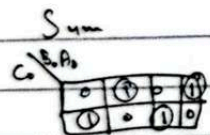
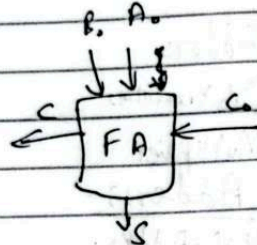
$$S = \bar{A}B + A\bar{B} = A \oplus B$$

$$C = AB$$

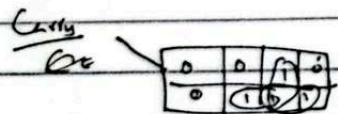


Full Adder:- adds 3 one bit numbers

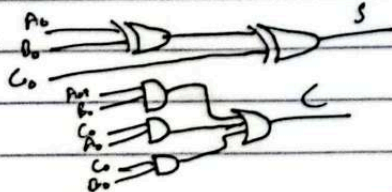
A	B	C	S
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1



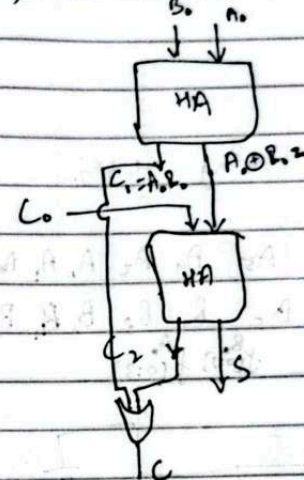
$$\begin{aligned}
 S &= \bar{C} \cdot \bar{B} \cdot A + \bar{C} \cdot B \cdot \bar{A} + C \cdot \bar{B} \cdot \bar{A} + C \cdot B \cdot A \\
 &= \bar{C} \cdot (\bar{B} \cdot A + B \cdot \bar{A}) + C \cdot (\bar{B} \cdot \bar{A} + B \cdot A) \\
 &= \bar{C} \cdot (A \oplus B) + C \cdot (A \oplus B) \\
 &= A \oplus B \oplus C
 \end{aligned}$$



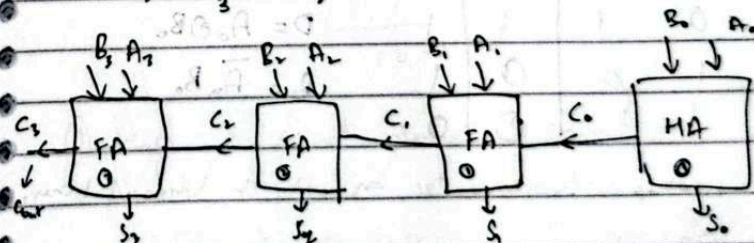
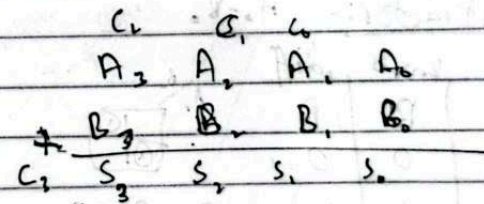
$$C = A \cdot B + C \cdot A + C \cdot B$$



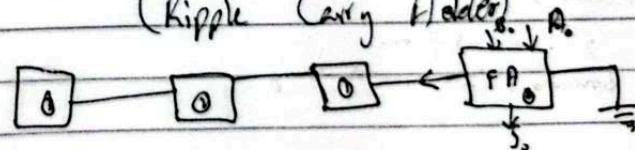
Q) Construct FA using HA.

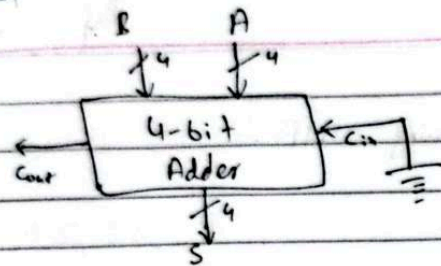


Q) Construct 4-bit adder using FA & HA.

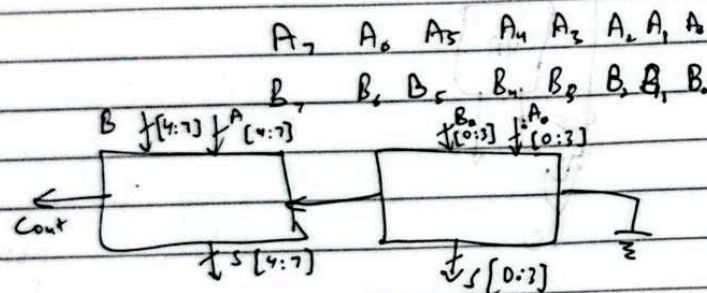


(Ripple Carry Adder)





Q) Construct 8-bit adder using 4-bit.



Subtractors:-

Half-Subtractor:- $A - B$.

		Borrow	Diff
A_0	B_0	B	D
0	0	0	0
0	1	1	1
1	0	0	1
1	1	0	0



$$D = A_0 \oplus B_0$$

$$B = \bar{A}_0 B_0$$

⇒ Arrange borrow 2 as it is base of binary.

0-1 x

Borrow 2

$$2-1 = 1 = \text{diff}$$

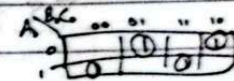
Borrow = 1 (2 in terms of binary)

Full-Subtractor:-

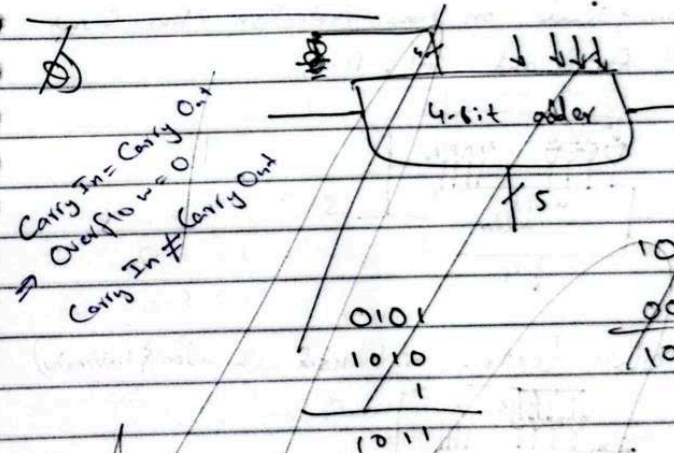
A_0	B_0	C_0	B	D
0	0	0	0	0
0	0	1	1	0
0	1	0	0	1
0	1	1	1	1
1	0	0	0	1
1	0	1	1	0
1	1	0	0	0
1	1	1	1	0



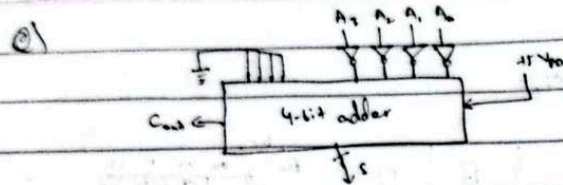
$$B = \bar{A}_0 C_0 + \bar{A}_0 B_0 + B_0 C_0$$



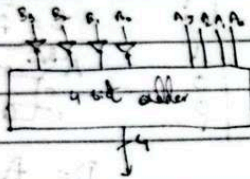
$$D = A_0 \oplus B_0 \oplus C_0$$



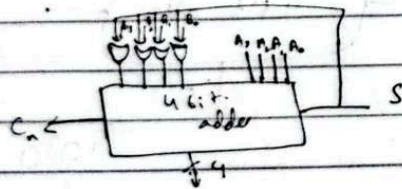
Q) Design a circuit that performs + - simultaneously based on some selection line as such that $S=0$, it performs $A+B$ else $S=1$, $A-B$



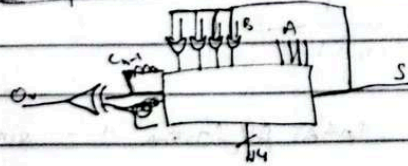
$$A - B = A + B = A + (-B) = A + (2^5 \oplus B)$$



Q) Design a circuit that performs addition & subtraction simultaneously based on some selection line S such that $S=0$, $A+B$ and $S=1$, $A-B$.



Overflow:- (extra bit used with adder & subtractor)



⇒ If the no. of bit in the sum are more than the individual number, ~~at that~~ we can say overflow occurred.

How to detect?

⇒ are in sign numbers

Example

Overflow bit 1
here as ans
is five

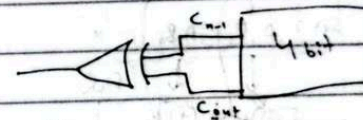
$$0 = 1, C = 1$$

1	1	0	1	13	-3
1	0	1	0	10	-6
1	0	1	1	25	-9

$C_{in} \neq C_{out}$
XOR

Left number
1 bit to
the number
is added
1 add number

If carry in of sign bit position and carry out of sign, same $\Rightarrow$ no overflow
different $\Rightarrow$ overflow



This hardware has capability to detect overflow

$$\begin{matrix} C_{in} & 0 & 0 & 0 \\ & 0 & 1 & 1 \\ & 0 & 1 & 1 \end{matrix}$$

$$\begin{matrix} & 0 & 1 & 0 \\ C_{out} & 0 & 1 & 0 \end{matrix} \Rightarrow \text{Over}$$

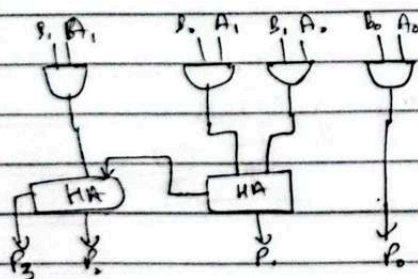
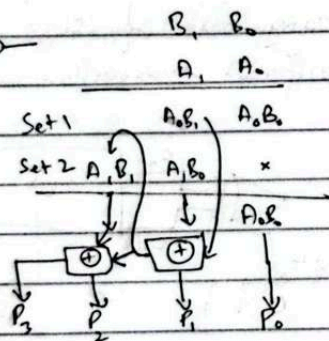
c) Multiplier:-

1-bit Multiplier

B_0	A_0	F
0	0	0
0	1	0
1	0	0
1	1	1

2 bit $\times$ 2 bit = 4 bit
 2 bit $\times$ 3 bit = 5 bit

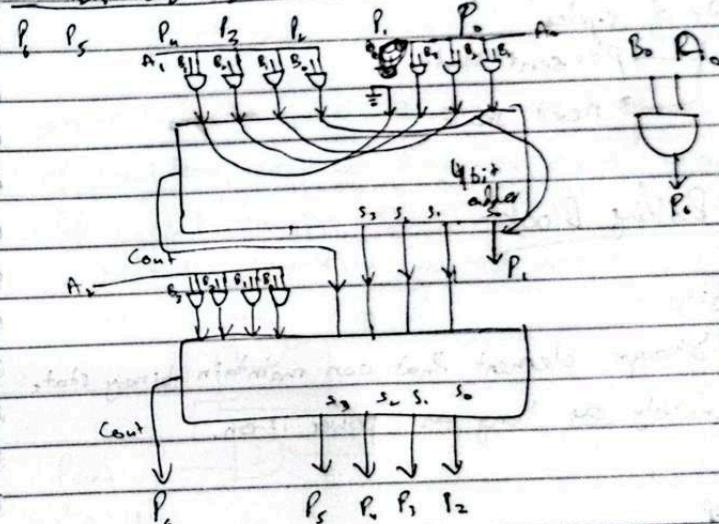
No. of set dependent
 on no. of bit of number



Full adder is
 required when
 carry from
 previous bit

4-Bit Multiplication:-

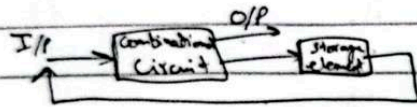
B_3	B_2	B_1	B_0
A_2	A_1	A_0	
A_3	A_2	A_1	A_0
A_3	A_2	A_1	A_0
A_3	A_2	A_1	A_0
A_3	A_2	A_1	A_0



B_3	B_2	B_1	B_0

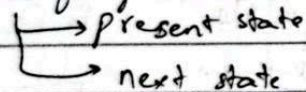
Sequential Circuits:-

↳ involve memory



Storage element → predicate present state

State of system



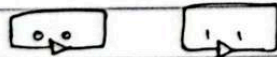
Basic Building Blocks:-

1) Latch:-

Storage element that can maintain binary state

indefinitely as long as power is on.

1) Flip Flop



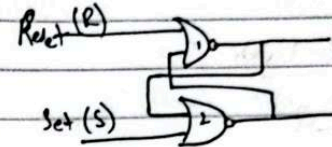
Set state → output equal to 1

Reset state → output equal to 0

Hold state/Preserve state → holds output as is

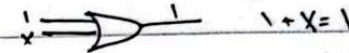
Latch stores 1 bit

1) SR Latch (NOR Latch)



S	R	Q	$\bar{Q}$	
(set state)	1 0	1 0		at
(hold state)	0 0	1 0		high
(reset state)	0 1	0 1		reset
	0 0	0 0		hold
	1 1	undefined		

↳ '1' dominance in OR gate



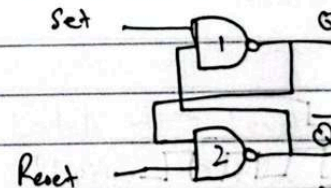
Q & Q̄ must be complement of each other

⇒ NOR latch ⇒ input 0 is never applied as input

⇒ Problem (done input 1 sai dono output 0 which is not true) ⇒ No solution

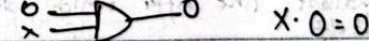
1) SR Latch (NAND Latch)

single set low krnai sai high output nikahe



S	R	Q	$\bar{Q}$	
0	1	1	0	(set)
1	1	1	0	(hold)
1	0	0	1	(reset)
1	1	0	1	(hold)
0	0			undefined

↳ '0' dominance

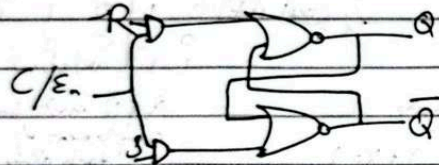


⇒ No control & No enable is in SR & SR Latch.

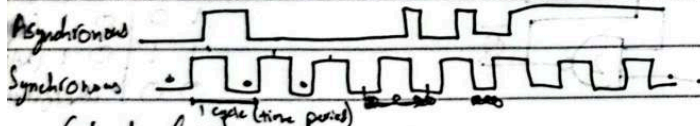
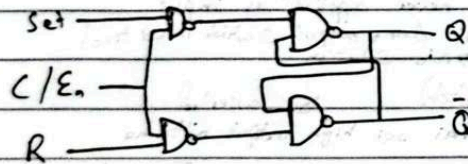
we has no control

Control Signal:-

1) Control Signal in NOR Latch



2) Control Signal in NAND Latch



Clock Cycle

Time period (smallest time after which signal repeats)

(unit) Frequency (cycles per second)

$$2.6 \text{ GHz} \Rightarrow 2.6 \times 10^9$$

Master Clock (sari device ka same hoga)

Clock (precise time gis ma hold, reset, set allow hoga)

→ wo point jis ma signal change ho raha hota ha

a) Positive edge / Rising edge : low to high transition

a) Negative / Falling edge : high to low transition

Clock Signal → Symmetric
→ periodic square wave

Clock ma synchronous circuit, not asynchronous
→ Components work simultaneously in clock

→ Clock (Zero) hold the data for complete time

→ Clock (One) set & reset the data

→ level sensitive

latch → level triggered (trigger → change state at clock)
(fluctuation chances nahi)

→ FlipFlops are edge triggered (set reset give time) (not complete)
(only for some time) (input is transfer to output for some time)

Sequential Circuit

→ input output

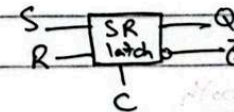
→ state variable

→ present state / next state
→ function of time $Q(t)$ $Q(t+1)$
Act

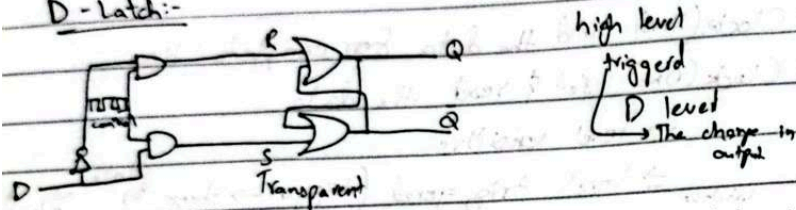
→ Set : output 1
→ Reset : output 0

Latch → Level triggered

FlipFlop → Edge triggered



D-Latch:-



→ Latches are transparent (jo input par apply karta hai wo output mil jata wo hold nhi krta)

→ High level triggered (clock ka +ve cycle me data transfer hota)

→ Low level triggered (clock ka -ve cycle me data transfer hota) (inverter use kr hai)

→ Latch : Synchronous & asynchronous both level triggered

→ FlipFlop : Synchronous & edge

Constant value → edge maintain



Flip Flop & lead at specific time instant

→ it rather lead at +ve

→ it rather lead at -ve

→ some system can read at both

Q → next state

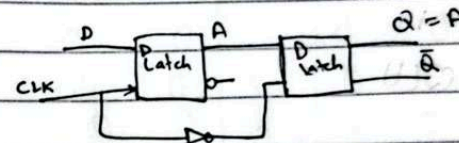
input is D
Control is 1.

C	D	Q(t+1)
0	X	Q(t) (hold)
1	0	0
1	1	1

C.O not enable
D → 0 hoga I aur na previous state ko retain krne

→ Latch half mei transparent hoti

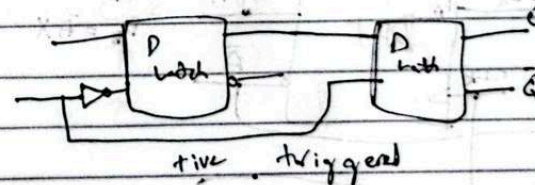
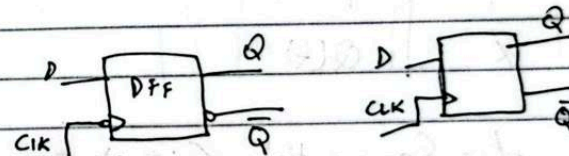
→ level triggered utilized in edge triggered



→ Low level triggered

Negative edge triggered flip flop

FlipFlop
clk active
data load
data enable
pehla dekhie



Master Slave Flip Flop

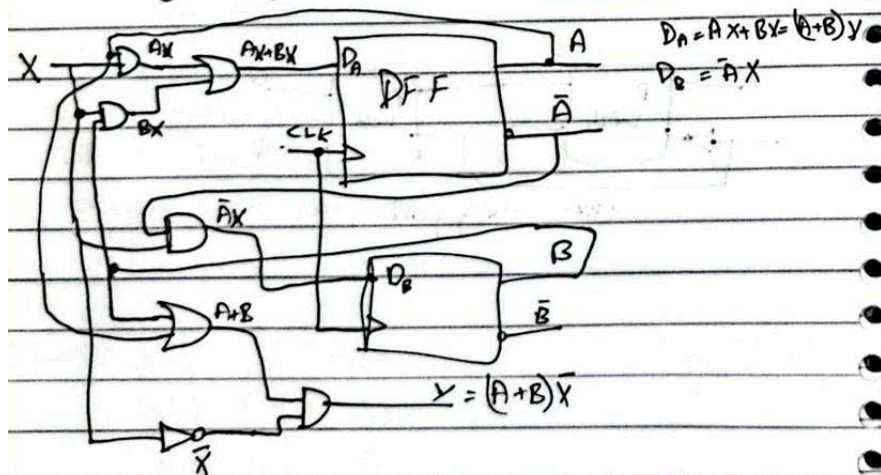
↓ -ive edge

CLK	D	$Q(t+1)$
↓	0	0
↓	1	1
	X	$Q(t)$

↑ +ive edge

CLK	D	$Q(t+1)$
↑	0	0
↑	1	1
	X	$Q(t)$

Analysis of Sequential Circuit



State Table :-

Present State		Input	Flip Flop Input		Next State		Output
Ax	Bx	X	D_A	D_B	$A(t+1)$	$B(t+1)$	Y
0	0	0	0	0	0	0	0
0	0	1	0	1	0	1	0
0	1	0	0	0	0	0	1
0	1	1	1	1	1	1	0
1	0	0	0	0	0	0	1
1	0	1	1	0	1	0	0
1	1	0	0	0	0	0	1
1	1	1	1	0	1	0	0

D Flipflop next state is equal to input

CLK	D	$Q(t+1)$
↑	0	0
↑	1	1

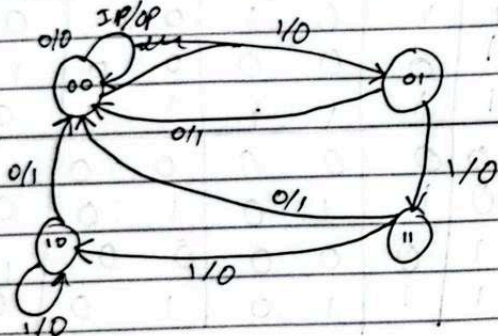
State Diagram:-

No. of Circle = No. of State

No. of States = 2^n

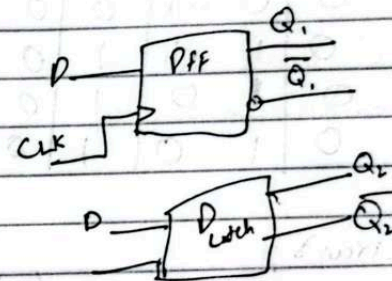
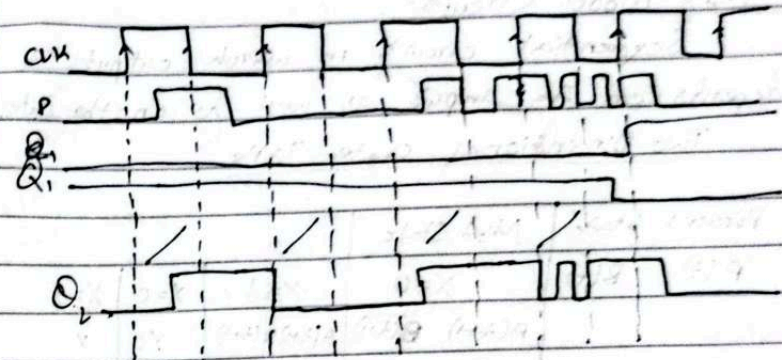
where n = no. of flip Flops

⇒ 1 circle ⇒ 2 lines emit possible input



Present State → Next state

Timing diagram



High Level Triggered

Latch transparent in +ive half cycle so +ive cycle me behavior hoga
fluctuation wise matter

Mealy Model Circuits-

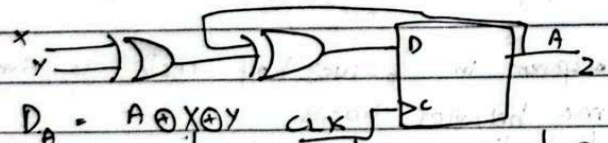
Sequential circuits in which output depends on the inputs as well as on the states

Two-Dimensional State Table

Present state		Next state					
A(t)	B(t)	X=0		X=1		X=0	
		A(t+1)	B(t+1)	A(t+1)	B(t+1)	Y	Y
0	0	0	0	0	1	0	0
0	1	0	0	1	1	1	0
1	0	0	0	1	0	1	0
1	1	0	0	1	0	1	0

Moore Model Circuit

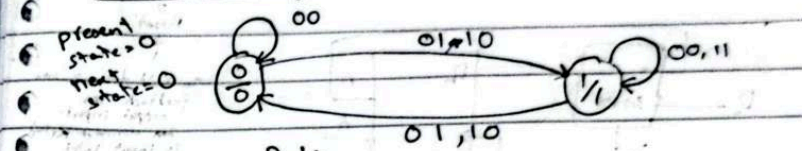
Output depends on states only



$D_A = A \oplus X \oplus Y$

Present State	Input		Next State	Output
A(t)	X	Y	$D_A = A(t+1)$	$Z = A$
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	0
1	0	0	1	1
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

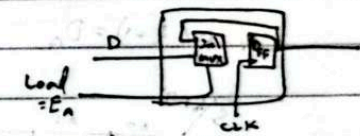
State Diagrams-



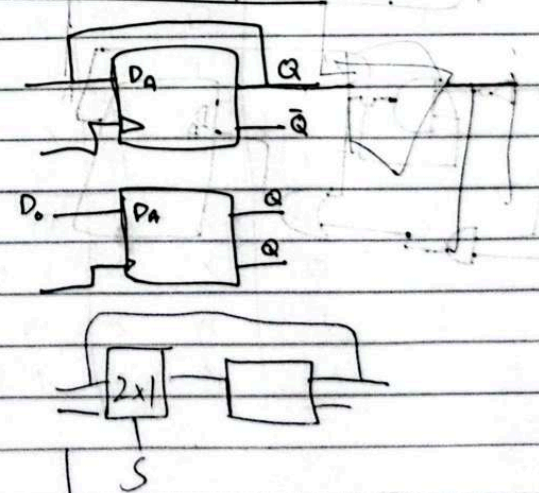
Design Problem

Design a digital system that has following characteristics:
1) It holds/preserves 1 bit of data for as long as user wants

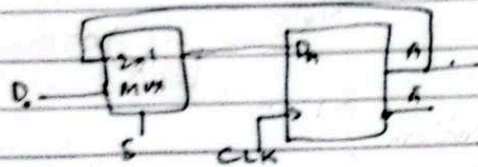
2) The user should have full control to set/reset the data whenever he/she wants (combination circuit)



1-bit Register with parallel load



Logic Design



Right
Clock
0
input
no valid data
no feedback
no output input
no external input
is input data
with output

Selection hardware → MUX

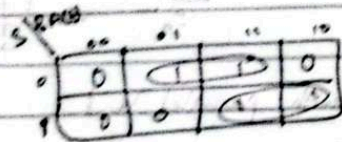
S=0 without state = Present state

State Table

S	Input D_n	Present state $A(t)$	Next state $A(t+1) = D_n$
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

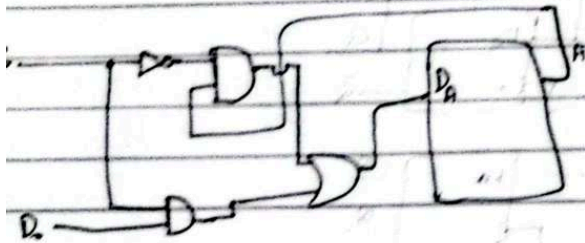
Data inputs
Present state
selected by
selection line

for $A(t+1) = D_n$

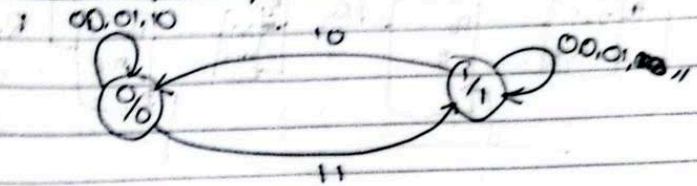


$$D_A = A(t+1)$$

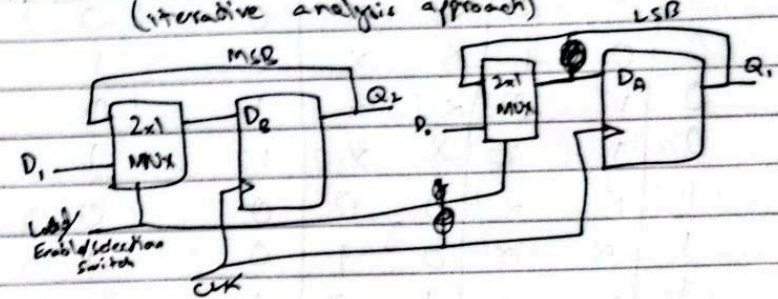
$$D_A = SA(t) + SD_n$$



State Diagram of 1-bit Register with Parallel load



2-bit Register with parallel load (iterative analysis approach)



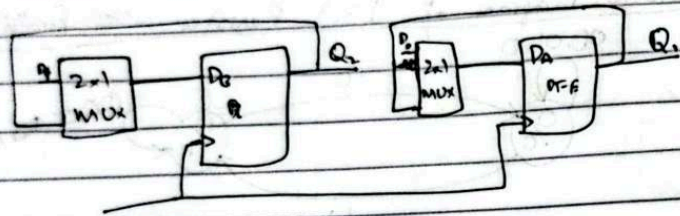
Compressed State Table :-

S	D ₁	D ₀	Q ₁ (t)	Q ₀ (t)	Q ₁ (t+1)	Q ₀ (t+1)
0	x	x	0	0	0	0
0	x	x	0	1	0	1
0	x	x	1	0	1	0
0	x	x	1	1	1	1
1	0	0	x	x	0	0
1	0	1	x	x	0	1
1	1	0	x	x	1	0
1	1	1	x	x	1	1

State S=0
decide state

S=1
data lines
give data

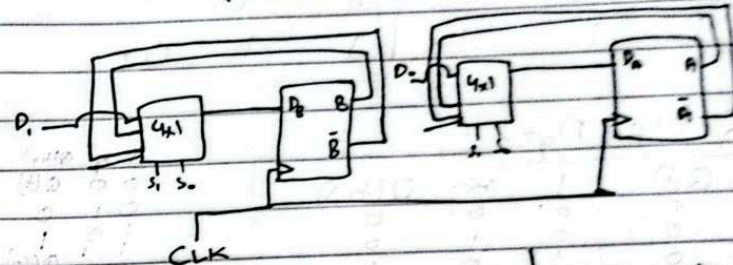
S=1
data lines
decides what
will output



S	P ₁	P ₀	Q ₁ (t)	Q ₂ (t)	Q ₁ (t+1)	Q ₂ (t+1)
0	0	0	x	x	0	0
0	0	1	x	x	0	1
0	1	0	x	x	1	0
0	1	1	x	x	1	1
1	x	x	0	0	0	0
1	x	x	0	1	0	1
1	x	x	1	0	1	0
1	x	x	1	1	1	1

Q) Design a digital system

- holds 2-bit of data
- loads 2-bit of data
- Toggles the data
previous value is complement



S ₁	S ₀	D ₁	D ₀	B(t)	A(t)	B(t+1)	A(t+1)
0	0	x	x	0	0	0	0
0	0	x	x	0	1	0	1
0	0	x	x	1	0	1	0
0	0	x	x	1	1	1	1
0	1	0	0	x	x	0	0
0	1	0	1	x	x	0	1
0	1	1	0	x	x	1	0
0	1	1	1	x	x	1	1
1	0	x	x	0	0	1	1
1	0	x	x	0	1	1	0
1	0	x	x	1	0	0	1
1	0	x	x	1	1	0	0
1	1	x	x	x	x	x	x
1	1	x	x	x	x	x	x
1	1	x	x	x	x	x	x
1	1	x	x	x	x	x	x

Characteristic Table:- (for analysis)

$Q(t)$	D	$Q(t+1)$
0	0	0
0	1	1
1	0	0
1	1	1

JK Flip Flop:-

$Q(t)$	J	K	$Q(t+1)$
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

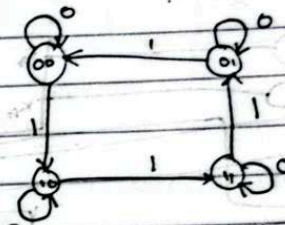
Excitation table:- (Design)

$Q(t)$	$Q(t+1)$	$D=Q(t+1)$
0	0	0
0	1	1
1	0	0
1	1	1

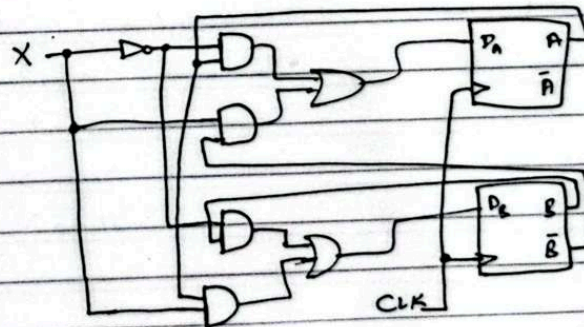
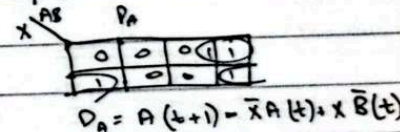
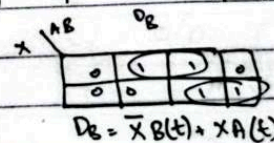
T Flip Flop:-

$Q(t)$	$Q(t+1)$	T
0	0	0
0	1	1
1	0	1
1	1	0

Q) Design a seq. circuit using 2-DFF and input X. X=0, the state of system remains unchanged. X=1, circuit goes through state transition from 00 $\rightarrow$ 01 $\rightarrow$ 11 $\rightarrow$ 10 and back to 00 and repeat.



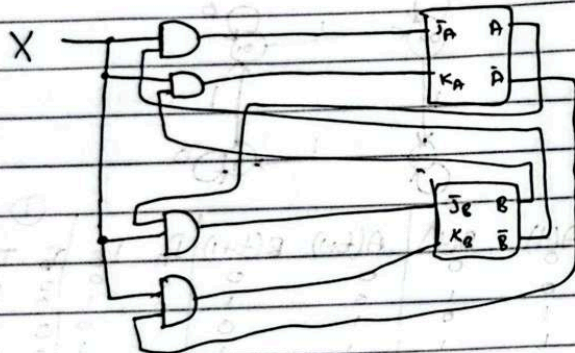
X	A(t)	B(t)	A(t+1)	B(t+1)	D_A	D_B
0	0	0	0	0	0	0
0	0	1	0	1	0	1
0	1	0	1	0	1	0
0	1	1	1	1	1	1
1	0	0	0	0	0	0
1	0	1	0	1	0	1
1	1	0	1	0	1	0
1	1	1	1	1	1	1





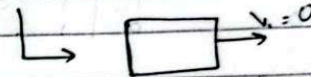
$$J_A = X\bar{B}(t) \quad J_B = XA(t)$$

$$K_A = XB(t) \quad K_B = X\bar{A}(t)$$



Design of Sequence Detector:-

10110011100110001
00000010000100001



3-bit sequence
001

S_0 = no bit has been received

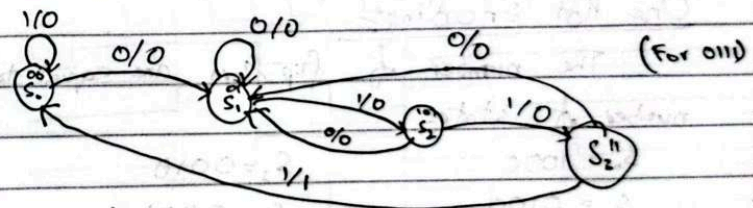
Assigning bits:-

S_0 = Initial State : No bit

S_1 = 1st bit has arrived

S_2 = 2nd bit has arrived

S_3 = 3rd bit has arrived



Assigning codes:-

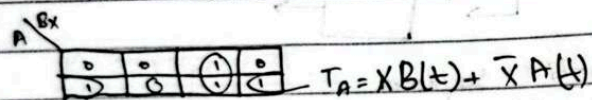
$S_0 = 00$

$S_1 = 01$

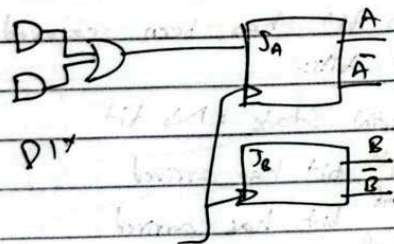
$S_2 = 10$

$S_3 = 11$

A(t)	B(t)	X	A(t+1)	B(t+1)	T _A	T _B	Y
0	0	0	0	1	0	0	0
0	0	1	0	0	0	0	0
0	1	0	1	0	0	0	0
0	1	1	0	0	1	0	0
1	0	0	0	1	0	1	0
1	0	1	0	0	1	1	0
1	1	0	1	1	0	1	0
1	1	1	1	1	0	1	0



Normal binary Encoding



One Hot Encoding:-

The number of flip flops are equal to the number of states.

$$S_0 = 1000$$

$$S_3 = 0010$$

$$S_1 = 0100$$

$$S_2 = 0001$$

Present State:-

Next State

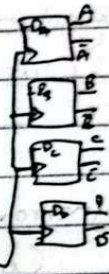
A(t)	B(t)	C	D	X=0	X=1
1	0	0	0	0100	1000
0	1	0	0	0100	0010
0	0	1	0	0100	0001
0	0	0	1	0100	1000

$$A(t+1) = D_A = AX + DX$$

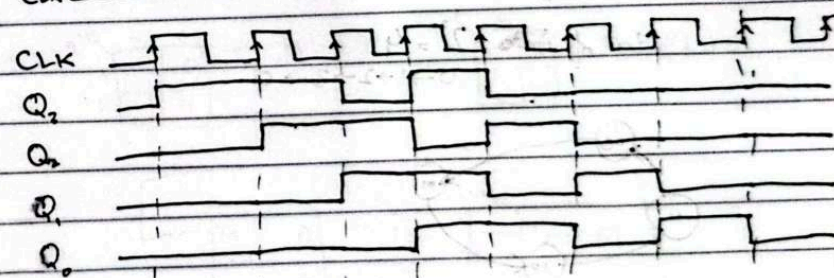
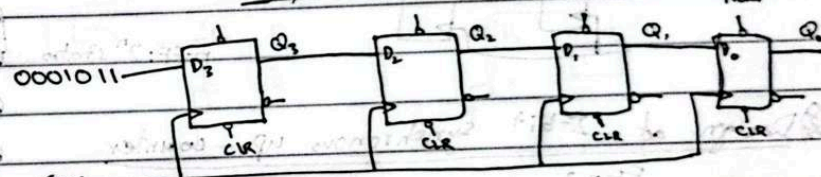
$$B(t+1) = A\bar{X} + B\bar{X} + C\bar{X} + D\bar{X} = (A+B+C+D)\bar{X}$$

$$C(t+1) = BX$$

$$D(t+1) = CX$$



Shift Register - SISO (serial in serial out)



Shift Right

SISO

SIPO

PIPO

PISO

S=Serial

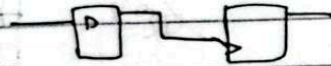
P=Parallel

Counters:-

Sequential circuits that count the number of clock pulses.

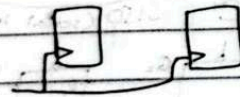
1) Asynchronous Counter:-

Output of 1-FF acts as clock for next



2) Synchronous Counters:-

Common clock for all FF



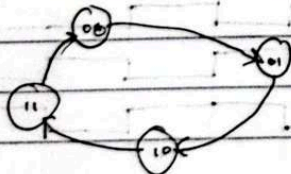
n-FF: 2^n states

Design of 2-bit synchronous up counter

FF=2

No. of states = $2^2 = 4$

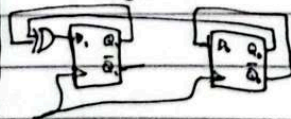
0 → 1 → 2 → 3 → 0



Present state	Next state	Next state
$Q_1(t)$	$Q_0(t)$	$Q_1(t+1)$ $Q_0(t+1)$
0	0	0 1
0	1	1 0
1	0	1 1
1	1	0 0

Q_1	Q_0	D_1	D_0
0	0	0	1
0	1	1	0
1	0	1	1
1	1	0	0

Q_1	Q_0	D_1	D_0
0	0	0	1
0	1	1	0
1	0	1	1
1	1	0	0

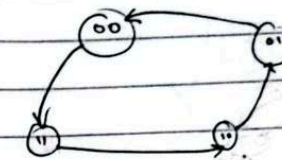


Design of 2-bit synchronous down counter

FF=2

No. of states = $2^2 = 4$

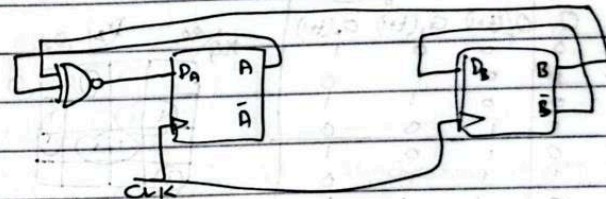
3 → 2 → 1 → 0



A	B	$A(t+1)$	$B(t+1)$	T_A	T_B
0	0	1	1	1	1
0	1	0	0	0	1
1	0	0	1	1	1
1	1	1	0	0	1

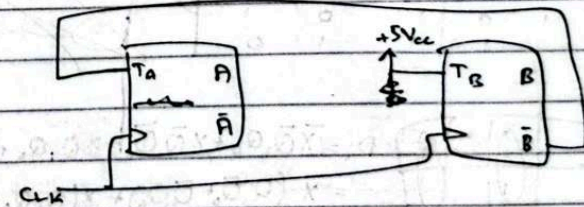
A	B	D_A	D_B
0	0	1	1
0	1	0	0
1	0	0	1
1	1	1	0

A	B	D_A	D_B
0	0	1	1
0	1	0	0
1	0	0	1
1	1	1	0



A	B	T_A	T_B
0	0	1	1
0	1	0	1
1	0	1	1
1	1	0	1

A	B	T_A	T_B
0	0	1	1
0	1	0	1
1	0	1	1
1	1	0	1

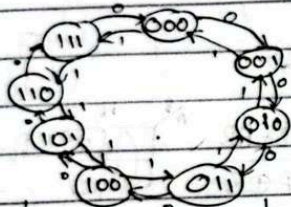


$$A = Q_1 \oplus Q_0$$

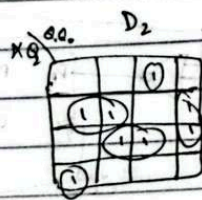
$$\bar{X}(A) = X(\bar{A})$$

$$X \oplus A = X \oplus Q_1 \oplus Q_0$$

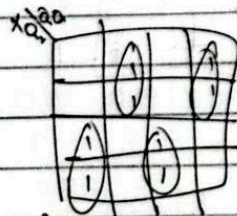
Q Design 3-bit synchronous updown counter with a control input X such that when $X=0$, it acts as upcounter and $X=1 \Rightarrow$ down counter.



X	Q_2	Q_1	Q_0	$Q_2(t+1)$	$Q_1(t+1)$	$Q_0(t+1)$
0	0	0	0	0	0	1
0	0	0	1	0	1	0
0	0	1	0	0	1	1
0	0	1	1	1	0	0
0	1	0	0	1	0	1
0	1	0	1	1	1	0
0	1	1	0	1	1	1
0	1	1	1	0	0	0
1	0	0	0	1	1	1
1	0	0	1	1	1	0
1	0	1	0	1	1	1
1	0	1	1	0	0	0
1	1	0	0	0	0	1
1	1	0	1	0	0	0
1	1	1	0	0	1	1
1	1	1	1	0	1	0



$$D_2 = \bar{X}\bar{Q}_2Q_1Q_0 + \bar{X}Q_2\bar{Q}_1\bar{Q}_0 + XQ_2Q_1Q_0 + XQ_2Q_1\bar{Q}_0 + XQ_2\bar{Q}_1Q_0 + XQ_2\bar{Q}_1\bar{Q}_0 + X\bar{Q}_2Q_1Q_0 + X\bar{Q}_2Q_1\bar{Q}_0$$



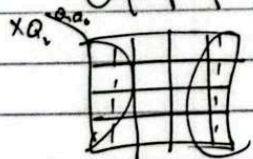
$$D_1 = \bar{X}\bar{Q}_2Q_1Q_0 + \bar{X}\bar{Q}_2Q_1\bar{Q}_0 + \bar{X}Q_2Q_1Q_0 + \bar{X}Q_2Q_1\bar{Q}_0 + X\bar{Q}_2Q_1Q_0 + X\bar{Q}_2Q_1\bar{Q}_0 + XQ_2\bar{Q}_1Q_0 + XQ_2\bar{Q}_1\bar{Q}_0$$

$$= \bar{X}(Q_2Q_1 + \bar{Q}_2Q_1) + X(Q_2\bar{Q}_1 + \bar{Q}_2Q_1)$$

$$= \bar{X}(Q_2 + \bar{Q}_2) + X(\bar{Q}_2 + Q_2)$$

$$= \bar{X} + X$$

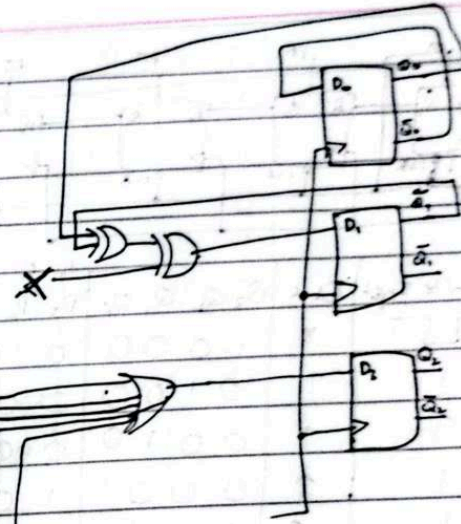
$$= 1$$



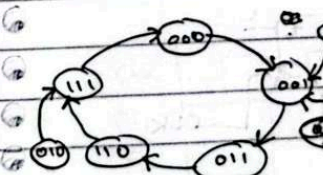
$$D_0 = \bar{Q}_0$$

$$30 \rightarrow 65\% \rightarrow 1$$

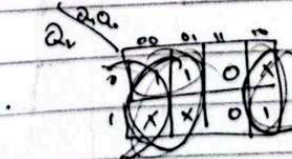
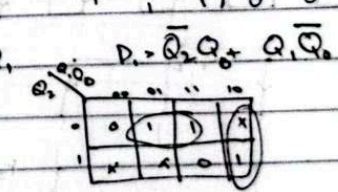
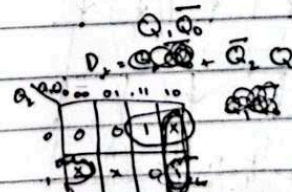
Forward Self Correction



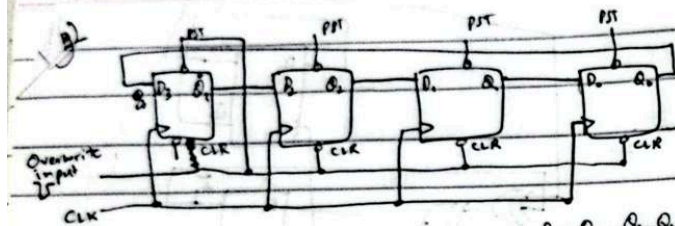
Q Design 3-bit synchronous counter that counts a random sequence 0,1,3,4,7 and repeat.



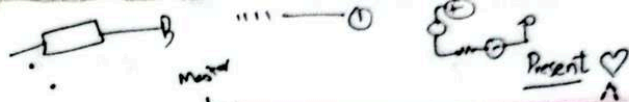
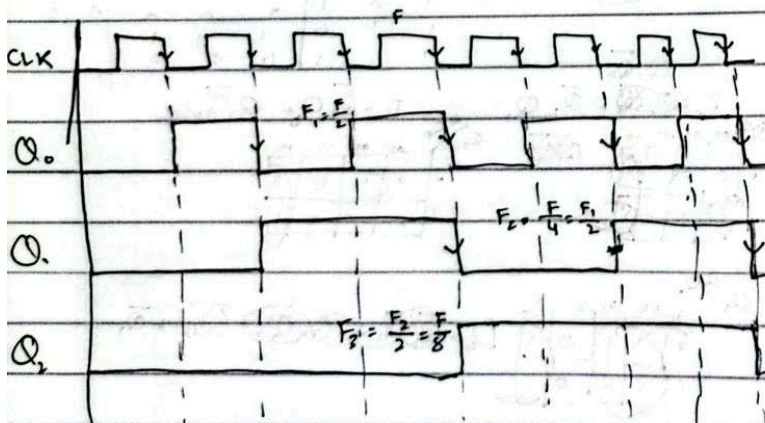
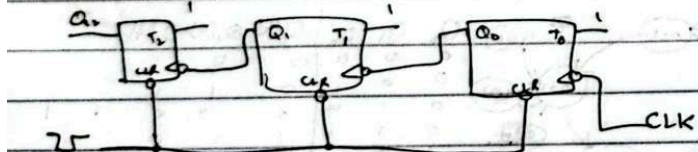
Q_2	Q_1	Q_0	$Q_2(t+1)$	$Q_1(t+1)$	$Q_0(t+1)$
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	0	1	1
0	1	1	1	0	0
1	0	0	1	0	1
1	0	1	1	1	0
1	1	0	1	1	1
1	1	1	0	0	0



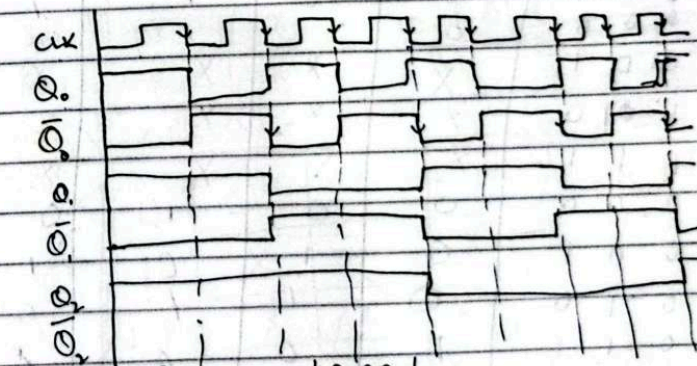
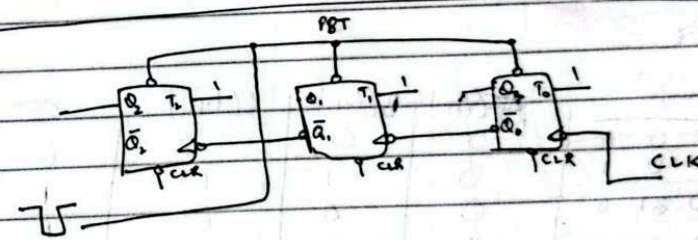
$$D_0 = \bar{Q}_0$$



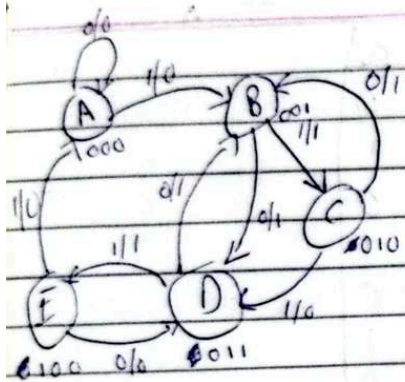
Overwrite Input	CLK	Q ₃	Q ₂	Q ₁	Q ₀	D ₃	D ₂	D ₁	D ₀
0	X	1	0	0	0	0	1	0	0
1	↑	0	1	0	0	0	0	1	0
1	↑	0	0	1	0	0	0	0	1
1	↑	0	0	0	1	1	0	0	0
1	↑	1	0	0	0	0	1	0	0
1	↑	0	1	0	0	0	0	1	0
1	↑	0	0	1	0	0	0	0	1



Overwrite input	CLK	Q ₂	Q ₁	Q ₀
X	↓	0	0	0
1	↓	0	0	1
1	↓	0	1	0
1	↓	0	1	1
1	↓	1	0	0
1	↓	1	0	1
1	↓	1	1	0
1	↓	1	1	1



Overwrite	CLK	Q ₂	Q ₁	Q ₀
X	↓	1	1	1
1	↓	1	1	0
1	↓	1	0	1
1	↓	1	0	0
1	↓	0	1	0
1	↓	0	0	1
1	↓	0	0	0



A B C D E

S	A	B	C	A(t+1)	B(t+1)	C(t+1)	Z
0	0	0	0	0	0	0	0
0	0	0	1	0	1	1	1
0	0	1	0	0	0	1	1
0	0	1	1	0	0	1	1
0	1	0	0	0	1	1	1
0	1	0	1	X	X	X	X
0	1	1	0	X	X	X	X
0	1	1	1	X	X	X	X
1	0	0	0	0	0	0	0
1	0	0	1	0	1	1	0
1	0	1	0	0	0	1	1
1	0	1	1	1	0	0	1
1	1	0	0	0	0	0	0
1	1	0	1	X	X	X	X
1	1	1	0	X	X	X	X
1	1	1	1	X	X	X	X

$\bar{T}_2$	$\bar{T}_1$	$\bar{T}$
0	0	0
0	1	0
0	0	1
0	1	0
1	1	1
X	X	X
X	X	X
X	X	X
0	0	1
0	1	1
0	0	1
1	0	1
1	1	1
X	X	X
X	X	X

$T_1 = C + \bar{S}A + B\bar{S}$

$\bar{T}_2$	$\bar{T}_1$	$\bar{T}$
0	0	0
0	1	0
0	0	1
0	1	0
1	1	1
X	X	X
X	X	X
X	X	X
0	0	1
0	1	1
0	0	1
1	0	1
1	1	1
X	X	X
X	X	X

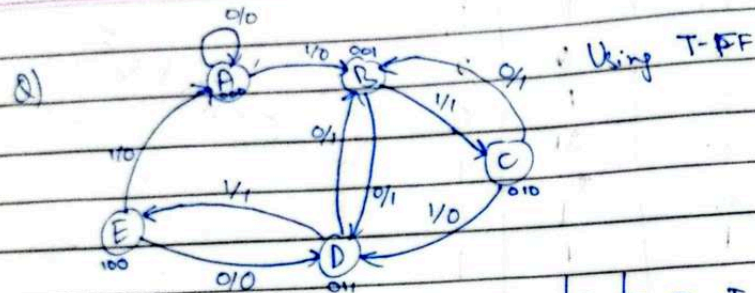
$\bar{T}_2$	$\bar{T}_1$	$\bar{T}$
0	0	0
0	1	0
0	0	1
0	1	0
1	1	1
X	X	X
X	X	X
X	X	X
0	0	1
0	1	1
0	0	1
1	0	1
1	1	1
X	X	X
X	X	X

$T_1 = SA + BCS + CS$

$\bar{T}_2$	$\bar{T}_1$	$\bar{T}$
0	0	0
0	1	0
0	0	1
0	1	0
1	1	1
X	X	X
X	X	X
X	X	X
0	0	1
0	1	1
0	0	1
1	0	1
1	1	1
X	X	X
X	X	X

$T_1 = SA + SA + X$

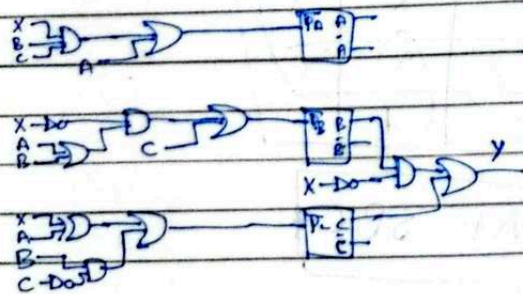
$T_2 = A + SB\bar{C}$



X	A	B	C	T _A (t+1)	T _B (t+1)	C(t+1)	D _{xy}	T _A	T _B	T _C
0	0	0	0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	0	0	0	0
0	0	1	0	0	0	0	0	0	0	0
0	0	1	1	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	0
0	1	0	1	0	0	0	0	0	0	0
0	1	1	0	0	0	0	0	0	0	0
0	1	1	1	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0	0
1	0	0	1	0	0	0	0	0	0	0
1	0	1	0	0	0	0	0	0	0	0
1	0	1	1	0	0	0	0	0	0	0
1	1	0	0	0	0	0	0	0	0	0
1	1	0	1	0	0	0	0	0	0	0
1	1	1	0	0	0	0	0	0	0	0
1	1	1	1	0	0	0	0	0	0	0

XA	BC	Y	XA	BC	Y	XA	BC	Y	XA	BC	Y
00	00	0	00	01	0	00	10	0	00	11	0
00	10	0	00	11	0	01	00	0	01	01	0
00	01	0	00	11	0	01	10	0	01	11	0
00	11	0	01	00	0	01	01	0	01	10	0
00	10	0	01	01	0	01	11	0	01	11	0
01	00	0	01	01	0	01	10	0	01	11	0
01	01	0	01	10	0	01	11	0	01	11	0
01	10	0	01	11	0	10	00	0	10	01	0
01	01	0	01	10	0	10	01	0	10	10	0
01	11	0	01	11	0	10	10	0	10	11	0
10	00	0	10	01	0	10	10	0	10	11	0
10	01	0	10	10	0	10	11	0	10	11	0
10	10	0	10	11	0	11	00	0	11	01	0
10	01	0	10	10	0	11	01	0	11	10	0
10	11	0	10	11	0	11	10	0	11	11	0
11	00	0	11	01	0	11	10	0	11	11	0
11	01	0	11	10	0	11	11	0	11	11	0
11	10	0	11	11	0						

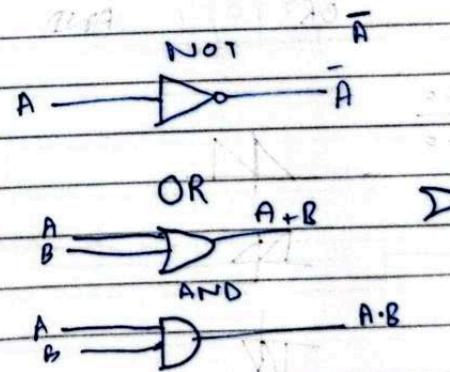
$T_A = X \oplus BC + A$
 $T_B = C + \bar{X}A + \bar{X}B = C + \bar{X}(A+B)$
 $T_C = \bar{X}A + X\bar{A} + \bar{B}C = (\bar{X} \oplus A) + \bar{B}C$
 $Y = C + \bar{X}B$



$$F = [(A + B\bar{C})(D + \bar{A}\bar{C})] + CD$$

$$\begin{aligned}
 &= (A + B)(A + \bar{C})(\bar{A} + D)(D + \bar{C}) \\
 &= (A + B\bar{C}) + (D + \bar{A}\bar{C}) + CD \\
 &= \bar{A}\bar{B}C + A\bar{C}\bar{D} + CD \\
 &= C(\bar{A}\bar{B} + A\bar{C}\bar{D} + D) \\
 &= C(\bar{A} + \bar{A}\bar{B})(\bar{A}\bar{B} + D) \\
 &= C(\bar{A} + \bar{A})(\bar{A} + \bar{B})(\bar{A}\bar{B} + D) \\
 &= C(\bar{A} + \bar{B})(\bar{A}\bar{B} + D) \\
 &= C(\bar{A}\bar{B} + (D + \bar{D})(\bar{A} + D)) \\
 &= C(\bar{A}\bar{B} + A + D) \\
 &= C(\bar{A} + \bar{B} + D) \\
 &= \bar{A}\bar{B} + A\bar{D} + C
 \end{aligned}$$

$$F1 = [(A + B\bar{C})(D + \bar{A}\bar{C})] + CD$$



A hand-drawn diagram of a cell, likely a plant cell, showing various organelles. The cell is bounded by a rectangular cell wall. Inside, there is a large central vacuole on the right side. A nucleus is located in the upper left. A mitochondrion is shown in the lower left. A Golgi apparatus is depicted in the center. A lysosome is shown in the lower right. The diagram is labeled with the following organelles: Nucleus, Mitochondrion, Golgi apparatus, Lysosome, and Vacuole.

~~1011~~

~~0100~~

~~818~~

~~101014~~

~~6~~

~~NOT~~

$$\frac{y_2 - 0}{x_2 - 0} = \frac{y - 0}{x - 0}$$
$$\frac{1}{2} = \frac{1}{2}$$

$\frac{2}{2}$ AND OR
 $\frac{2}{2}$ OR AND

The first diagram shows a Zener diode symbol with a horizontal line and a vertical line, labeled $Z-A$ and 10 .

The second diagram shows a Schottky diode symbol with a horizontal line and a vertical line, labeled $2-A$ and $0.4A$.

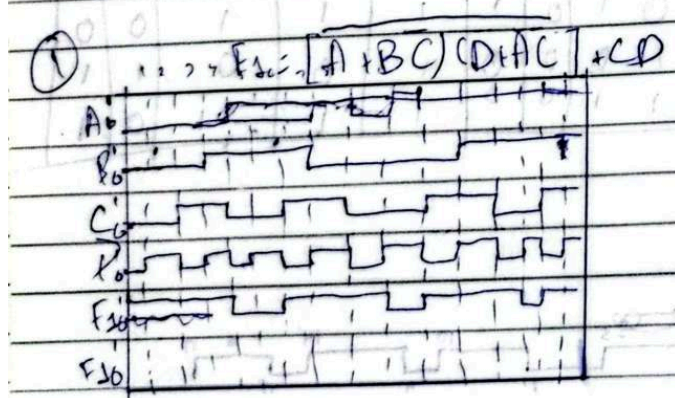
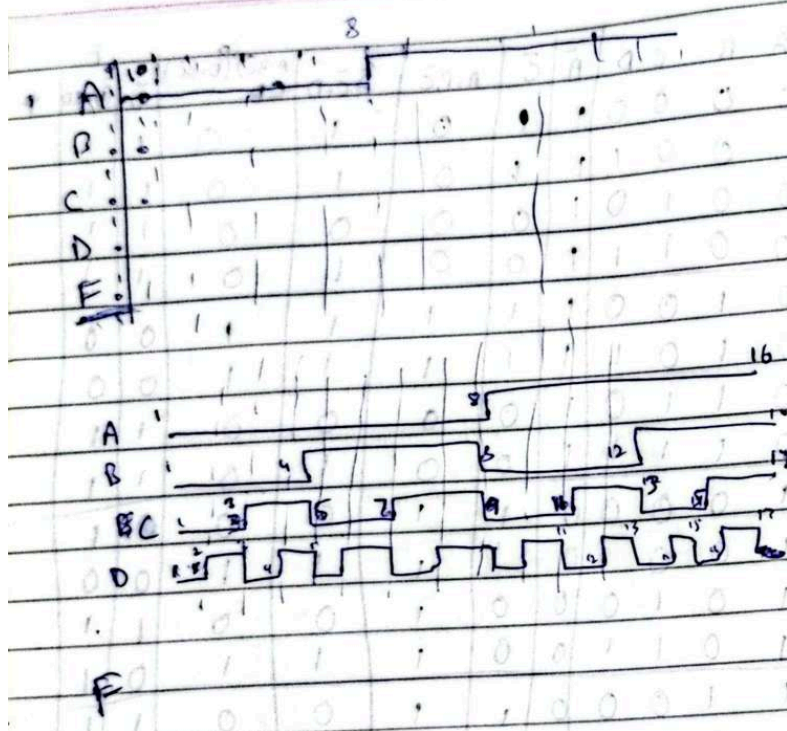
The third diagram shows an LED diode symbol with a horizontal line and a vertical line, labeled $2-A$ and $0.4A$.

$$16 = \frac{1}{4}$$

①										$(A \cdot B \cdot C) \vee (D \cdot A \cdot C)$		F	
A	B	C	D	$\bar{A}$	$\bar{C}$	$A \cdot B \cdot \bar{C}$	$\bar{A} \cdot \bar{C} \cdot D$	$(A \cdot B \cdot C) \vee (D \cdot A \cdot C)$	$\bar{F}$	$\bar{F} \oplus CD$	$\bar{F}$		
0	0	0	0	1	1	0	1	0	1	1			
0	0	0	1	1	1	0	1	0	1	1			
0	0	1	0	1	0	0	0	0	1	1			
0	0	1	1	1	0	0	1	0	1	1			
0	1	0	0	1	1	1	1	1	0	0			
0	1	0	1	1	1	1	1	1	0	0			
0	1	1	0	1	0	0	0	0	1	1			
0	1	1	1	1	0	0	1	0	1	1			
1	0	0	0	0	1	1	0	0	0	1	F		
1	0	0	1	0	1	1	1	1	0	0			
1	0	1	0	0	0	1	0	0	1	1			
1	0	1	1	0	0	1	1	1	0	1			
1	1	0	0	0	1	1	0	0	1	1			
1	1	0	1	0	1	1	1	1	0	0			
1	1	1	0	0	0	1	0	0	1	1			
1	1	1	1	0	0	1	0	0	1	1			
1	1	1	1	0	0	1	1	1	0	1			

①

[illegible]



②

SOP

$$G_1(2) = \bar{A}\bar{B}C\bar{D}$$

$$G_2(4,11) = B\bar{C}\bar{D}$$

$$G_3(12,14) = AB\bar{D}$$

$$F = \bar{A}\bar{B}C\bar{D} + B\bar{C}\bar{D} + AB\bar{D}$$

NOT 4
OR 3
AND 3
Gates 14

AB \ CD	00	01	11	10
00				1
01	1	1		
11	1			1
10				

A	B	C	D	$\bar{A}$	$\bar{B}$	$\bar{C}$	$\bar{D}$	$\bar{A}\bar{B}C\bar{D}$	$B\bar{C}\bar{D}$	$AB\bar{D}$	F
0	0	0	0	1	1	1	1	0	0	0	
0	0	0	1	1	1	1	0	0	0	0	
0	0	1	0	1	1	0	1	1	0	0	1
0	0	1	1	1	1	0	0	0	0	0	
0	1	0	0	1	0	1	1	0	1	0	1
0	1	0	1	1	0	1	0	0	0	0	
0	1	1	0	1	0	0	1	0	0	0	
0	1	1	1	1	0	0	0	0	0	0	
1	0	0	0	0	1	1	1	0	0	0	
1	0	0	1	0	1	1	0	0	0	0	
1	0	1	0	0	1	0	1	0	0	0	
1	0	1	1	0	1	0	0	0	0	0	
1	1	0	0	0	0	1	1	0	1	1	1
1	1	0	1	0	0	1	0	0	0	0	
1	1	1	0	0	0	0	1	0	0	1	1
1	1	1	1	0	0	0	0	0	0	0	

(POS)

AB \ CD	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	0	0	0	0
10	0	0	0	0

$$G_1(0,1,2,3) = A + B$$

$$G_2(0,4,8,12) = C + D$$

$$G_3(10,11,14,15) = \bar{A} + \bar{C}$$

$$G_4(0,1,4,5) = A + C$$

$$F = (A+B)(C+D)(\bar{A}+\bar{C})(A+C)$$

NOT $\Rightarrow 2$

OR $\Rightarrow 4$

AND $\Rightarrow 4$

Gates $\Rightarrow 10$

A(0) & B(1) & (B(0) || C(1))

||

A(1) & D(1) & (C(0) || D(0))

$$\Rightarrow \bar{A}B(\bar{B}+C) + AD(\bar{C}+\bar{D})$$

MUX
NOT $\Rightarrow 4$
OR $\Rightarrow 3$
AND $\Rightarrow 4$
Gates $\Rightarrow 11$

1

A	B	C	D	A'	B'	D'	A'BCD	A'B'CD	AB'CD	ABCD'	ABCD	F
0	0	0	0	1	1	1	0	0	0	0	0	0
0	0	0	1	1	1	0	0	0	0	0	0	0
0	0	1	0	1	1	1	0	0	0	0	0	0
0	0	1	1	1	1	0	0	0	0	0	0	0
0	1	0	0	1	0	1	0	0	0	0	0	0
0	1	0	1	1	0	0	0	0	0	0	0	0
0	1	1	0	1	0	1	0	0	0	0	0	0
0	1	1	1	1	0	0	0	0	0	0	0	0
1	0	0	0	0	1	1	0	0	0	0	0	0
1	0	0	1	0	1	0	0	0	0	0	0	0
1	0	1	0	0	1	1	0	0	0	0	0	0
1	0	1	1	0	1	0	0	0	0	0	0	0
1	1	0	0	0	0	1	0	0	0	0	0	0
1	1	0	1	0	0	0	0	0	0	0	0	0
1	1	1	0	0	0	1	0	0	0	0	0	0
1	1	1	1	0	0	0	0	0	0	0	0	0

(SOP)

AB \ CD	00	01	11	10
00				
01			1	1
11		1		
10		1		

$$G_1(6,7) = \bar{A}BC$$

$$G_2(9,13) = A\bar{C}D$$

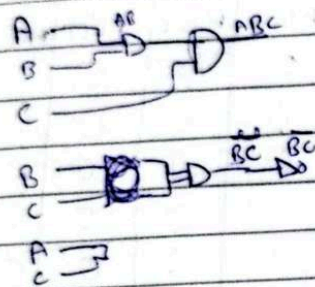
$$F = \bar{A}BC + A\bar{C}D$$

NOT $\Rightarrow 2$

OR $\Rightarrow 1$

AND $\Rightarrow 4$

Gates $\Rightarrow 7$



$$ABC + \bar{B}C + \bar{A}C$$

$$ABC + \bar{B} + \bar{C} + \bar{A} + \bar{C}$$

$$(\bar{B} + \bar{C})(\bar{B} + \bar{A} + \bar{C}) = \bar{A} + \bar{C}$$

$$\bar{B} + AC + \bar{A} + \bar{C}$$

$$\bar{B} + (\bar{A} + A)(\bar{A} + C) + \bar{C}$$

$$\bar{B} + \bar{A} + C + \bar{C}$$

$$\bar{B} + \bar{A}$$

⑥

SOP

$$E = \bar{A}\bar{B}\bar{C} + \bar{A}A\bar{B}\bar{C} + A\bar{A}B\bar{C} + A\bar{A}\bar{B}C$$

$$G = \bar{A}\bar{B}\bar{C} + A\bar{A}\bar{B}\bar{C} + A\bar{A}B\bar{C}$$

$$L = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + \bar{A}\bar{B}C$$

$$E = (A \text{ XNOR } B)(B \text{ XNOR } C)$$

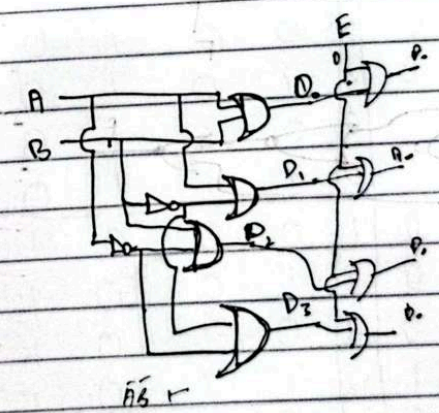
⑦

A	B	D ₁	D ₂	D ₃	D ₄	E
0	0	0	1	1	1	
0	1	1	0	1	1	
1	0	1	1	0	1	
1	1	1	1	1	0	

L	E	G
1	0	0
0	0	1

$E = 1 (\bar{L} + \bar{G})$
 $E = 0 (L + G)$

Active High = SOP (output 1)
Active Low = POS (output 0)

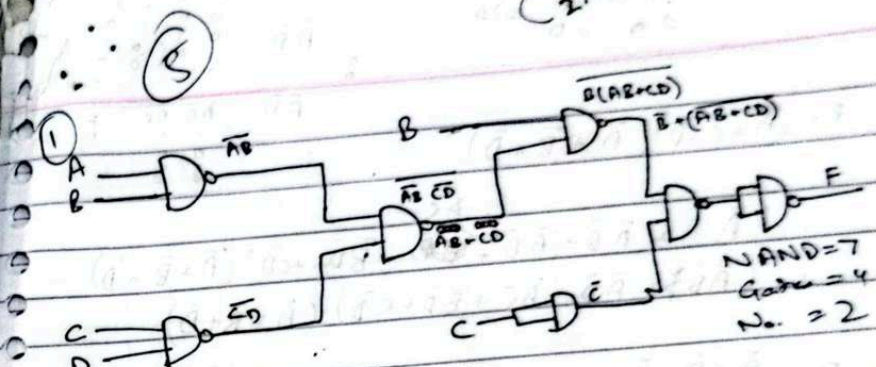
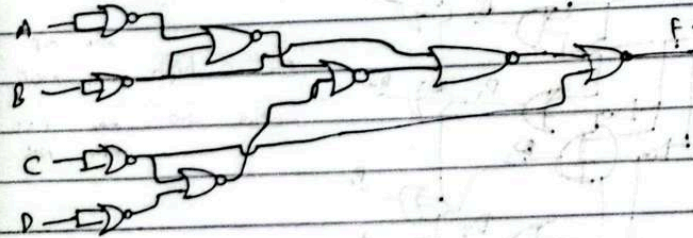
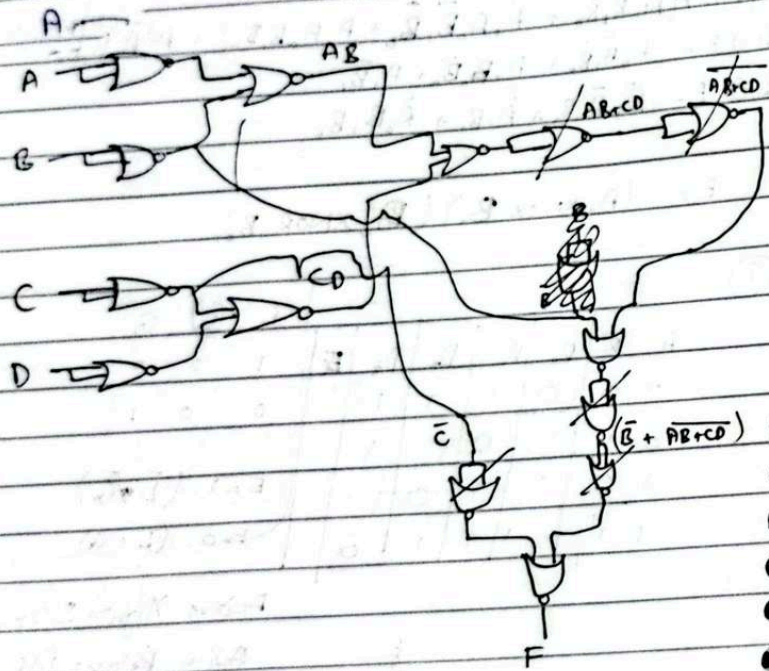


$$(A+B)(A+B)(A+B)(A+B)$$

min OR
Active high out NOR
Active low out AND
min NOT

SOP
 $\bar{A}\bar{C}$
 $\bar{B}\bar{C}$

00	01	11	10	00	00
01	01	11	00	00	00
11	00	00	00	00	00
10	11	00	00	00	00



① $F = (\bar{B} + AB + CD)\bar{C}$

A	B	C	D	$\bar{B}$	$AB + CD$	$\bar{B} + \bar{B}$	$(\bar{B} + \bar{B})\bar{C}$
0	0	0	0	1	0	1	1
0	0	0	1	1	0	1	1
0	0	1	0	1	0	1	0
0	0	1	1	1	1	0	0
0	1	0	0	0	0	1	1
0	1	0	1	0	0	1	0
0	1	1	0	0	1	0	0
0	1	1	1	0	1	0	0
1	0	0	0	1	0	1	1
1	0	0	1	1	0	1	1
1	0	1	0	1	0	1	0
1	0	1	1	1	1	0	0
1	1	0	0	0	1	0	0
1	1	0	1	0	1	0	0
1	1	1	0	0	1	0	0
1	1	1	1	0	1	0	0

$A=0$
 $\bar{A}\bar{D}$
 $\bar{A}\bar{D}$ $\bar{A}\bar{D}$ $\bar{A}\bar{D}$ F

$$= \bar{A}\bar{A} + \bar{A}\bar{B} + \bar{A}\bar{C} + \bar{A}\bar{B}\bar{C} + \bar{B}\bar{D} + \bar{C}\bar{D} (\bar{A} + \bar{B} + \bar{D})$$

$$= (\bar{A}\bar{B} + \bar{A}\bar{C} + \bar{B}\bar{D} + \bar{C}\bar{D}) (\bar{A} + \bar{B} + \bar{D})$$

$$xy + \bar{x}z + yz = xy + \bar{x}z$$

$$= \bar{A} + \bar{D} + \bar{B}C + \bar{A}\bar{B}D$$

$$(A + C + D)(\bar{A} + \bar{B} + \bar{D})$$

$$\bar{A}\bar{A} + \bar{A}\bar{B} + \bar{A}\bar{D} = \bar{A}\bar{C} + \bar{B}\bar{C} + \bar{C}\bar{D} + \bar{A}\bar{D} + \bar{B}\bar{D} + \bar{D}\bar{D}$$

$$\bar{A} + \bar{A}\bar{B} + \bar{A}\bar{D} = (\bar{A} + \bar{B} + \bar{D})C + \bar{A}\bar{D} + \bar{B}\bar{D} + \bar{D}$$

$$\bar{A}(1 + \bar{B} + \bar{D}) + (\bar{A} + \bar{B} + \bar{D})C + (\bar{B} + 1)\bar{D}$$

$$\bar{A}(1) + (\bar{A} + \bar{B} + \bar{D})(C + 1)D$$

$$\bar{A} + (\bar{A} + \bar{B} + \bar{D})(C + \bar{D})$$

NOT = 3
OR = 4
AND = 1

$$= \bar{A} \cdot \bar{B}C \cdot \bar{B}\bar{D} + \bar{D} \cdot \bar{A} \cdot \bar{B}C \cdot (C + \bar{B} + 1) \bar{D}$$

$$= \bar{A} * \bar{B} C + (\bar{B} + 1) \bar{D}$$

$$= \bar{A} + B\bar{C} + \bar{D}$$

$$A + (B + \bar{D})C + B$$

$$\cdot \bar{A} + B(C + \bar{D}) + \bar{C} + (C + 1)\bar{D}$$

$$\Rightarrow \bar{A} + \bar{B}(C + D) + \bar{D}$$

NOT = 3
OR = 3
AND = 1

$$AD = 1$$

21

2

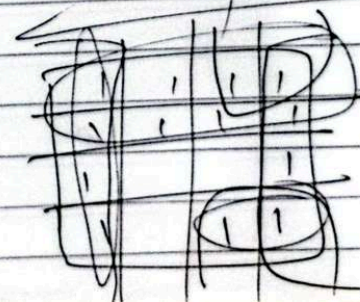
$A + C + D$	$\bar{A} + \bar{B} + \bar{D}$	F
1	1	1
1	1	1
1	1	1
1	1	1
1	1	1
1	1	1
1	1	1
1	1	1
1	1	1
0 1	1	1
0 0	1	0
0 1	1	1
0 0 1	1	1
0 1	1	1
0 0	0	0
0 1	1	1
0 0 1	0	0

$$(\bar{A} + \bar{D}) \cdot (1 + C) \cdot$$

$$(\bar{A} + C + \bar{D})(\bar{A} + \bar{B} + \bar{D})$$

$$(\bar{A} + \bar{D}) \left[\binom{1}{C} \binom{1}{\bar{B}} \right]$$

15.



POS

$G_1(0,1,8,9) = B + C$
 $G_2(8,9,10,11) = \bar{A} + B$
 $G_3(6,7) = A + \bar{B} + \bar{C}$
 $G_4(1) = \bar{D}$

AB \ CD	00	01	11	10
00	0	0	0	0
01	0	0	0	0
10	0	0	0	0
11	0	0	0	0

$F = (B+C)(\bar{A}+B)(A+\bar{B}+\bar{C})(\bar{D})$

NOT $\Rightarrow 4$
 OR $\Rightarrow 4$
 AND $\Rightarrow 3$
 Gate $\Rightarrow 11$

MIX

$F = \bar{A}\bar{B}\bar{C}\bar{D} + B\bar{C}\bar{D} + A\bar{B}\bar{D}$
 $= \bar{A}\bar{B}\bar{C}\bar{D} + \bar{B}\bar{D}(\bar{C} + A)$

NOT $\Rightarrow 4$
 OR $\Rightarrow 2$
 AND $\Rightarrow 5$
 Gate $\Rightarrow 11$

$\bar{C} = \bar{C}$
 $\bar{A} + B$
 $B + C$
 $A + \bar{B} + \bar{C}$

0	0	0	0
0	0	0	0
0	0	0	0
0	0	0	0

$n \times 2^n$

$2^n =$

Encoding

G
 B

P O B G

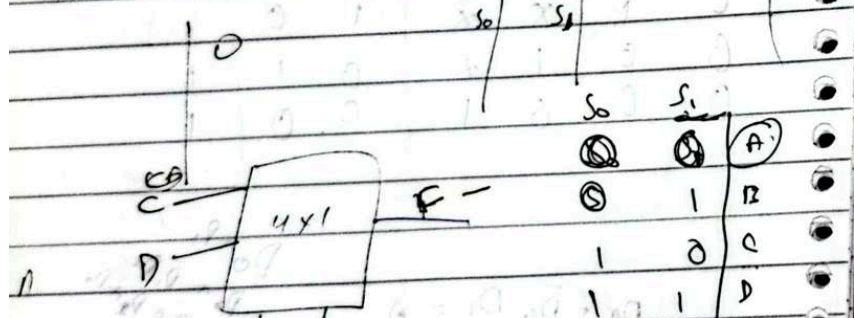
D_0	D_1	D_2	D_3	A	B	ΣV
0	0	0	0	x	x	0
1	x	x	x	1	1	1
0	1	x	x	1	0	1
0	0	1	x	0	1	1
0	0	0	1	0	0	1

$D_0 + \bar{D}_0 D_1 = A$
 $\bar{D}_0 \bar{D}_1 D_2 + D_0 = B$
 $D_0 + D_1 + D_2 + D_3 = V$

Don't
 Don't
 $B + D_2 + D_3 + D_0$

MUX

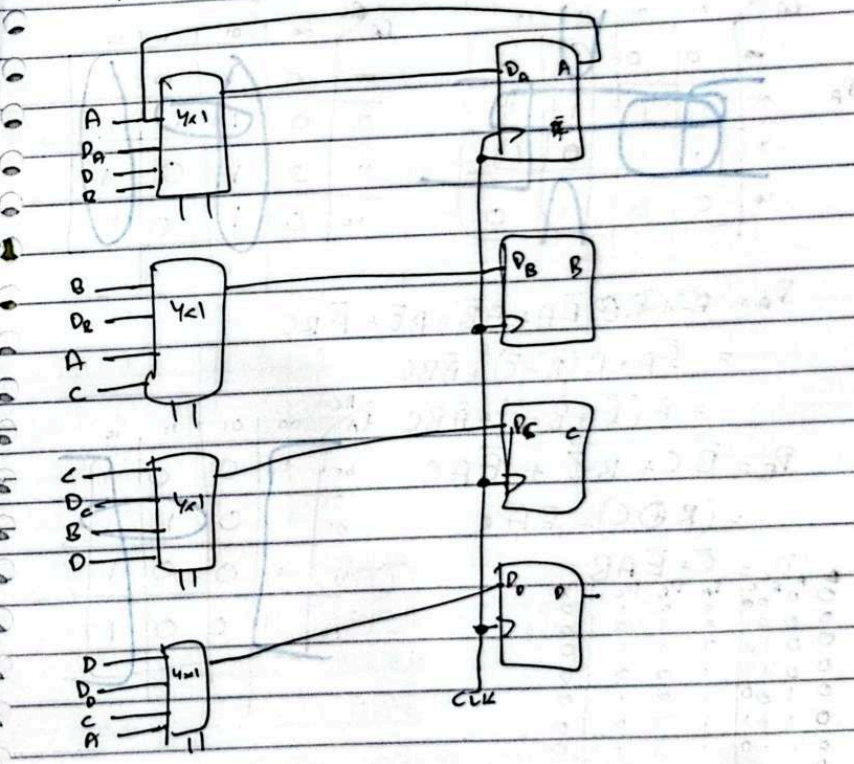
	A	B	C	D	Y	E	A'	B'	F
0	0	0	0	0		0	0	0	
1	0	0	1	0		0	0	1	
2	0	1	0	0		0	1	0	
3	0	1	1	0		0	1	1	
4	1	0	0	0		1	0	0	
5	1	0	1	0		1	0	1	
6	1	1	0	0		1	1	0	
7	1	1	1	0		1	1	1	



X	A	B	Y	E	F
00	0	0	0	0	0
01	0	1	0	0	1
10	1	0	0	1	0
11	1	1	0	1	1

$$F = \bar{S}_0 \bar{S}_1 A + \bar{S}_0 S_1 B + S_0 \bar{S}_1 C + S_0 S_1 D$$

S ₁	S ₀	D _A	D _B	D _C	D _D
0	0	A	B	C	D
0	1	D	A	B	C
1	0	B	C	D	A
1	1	C	D	A	B



y

EA \ BC	00	01	11	10
00	0	0	0	0
01	0	0	1	0
11	0	0	0	0
10	1	0	0	0

$D_A \leftarrow D_B \leftarrow D_C$

$$Y = E \bar{A} \bar{B} \bar{C} + \bar{E} A B C$$

D_A

EA \ BC	00	01	11	10
00	0	0	1	0
01	1	1	1	1
11	1	1	0	1
10	0	0	1	0

D_B

EA \ BC	00	01	11	10
00	0	0	0	1
01	0	1	1	1
11	0	1	0	1
10	0	1	0	1

$$D_A = \bar{E}A + A\bar{B} + A\bar{C} + \bar{A}BC$$

$$= \bar{E}A + A(\bar{B} + \bar{C}) + \bar{A}BC$$

$$= A(\bar{E} + \bar{B} + \bar{C}) + \bar{A}BC$$

$$D_B = \bar{B}C + B\bar{C} + \bar{E}AC$$

$$= (B \oplus C) + \bar{E}AC$$

$$D_C = \bar{C} + \bar{E}AB$$

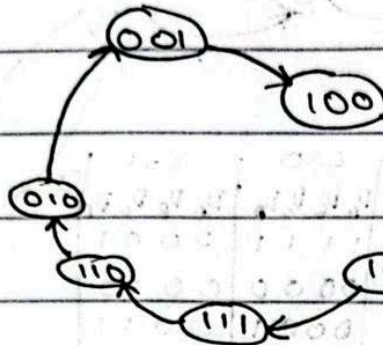
EA \ BC	00	01	11	10
00	1	0	0	1
01	1	0	1	1
11	1	0	0	1
10	1	0	0	1

E	A	B	C	D_A	D_B	D_C	Y
0	0	0	0	0	0	1	0
0	0	0	1	0	0	0	0
0	0	1	0	0	0	0	0
0	0	1	1	1	0	0	0
0	1	0	0	1	0	1	0
0	1	0	1	1	1	0	0
0	1	1	0	1	1	1	0
0	1	1	1	1	1	1	1
1	0	0	0	0	0	1	0
1	0	0	1	0	1	0	0
1	0	1	0	0	1	1	0
1	0	1	1	1	0	0	0
1	1	0	0	1	0	1	0
1	1	0	1	1	1	0	0
1	1	1	0	1	1	1	0
1	1	1	1	0	0	0	1

Q(x) Q(t+1) J K

0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

0 000
1 001
2 010
3 011
4 100
5 101
6 110
7 111



A B C	A(t+1)	B(t+1)	C(t+1)
0 0 0	X	X	X
0 0 1	1	0	0
0 1 0	0	0	1
0 1 1	1	0	1
1 0 0	0	1	1
1 0 1	1	1	1
1 1 0	0	1	0
1 1 1	1	1	0

J _A	K _A	J _B	K _B	J _C	K _C
X	X	X	X	X	X
1	X	0	X	X	1
0	X	X	1	1	X
1	X	X	1	X	0
X	1	1	X	1	X
X	0	1	X	X	0
X	1	X	0	0	X
X	0	X	0	X	1

A \ BC	00	01	11	10
0	X	1	1	0
1	0	1	1	0

$$A(t+1) = C$$

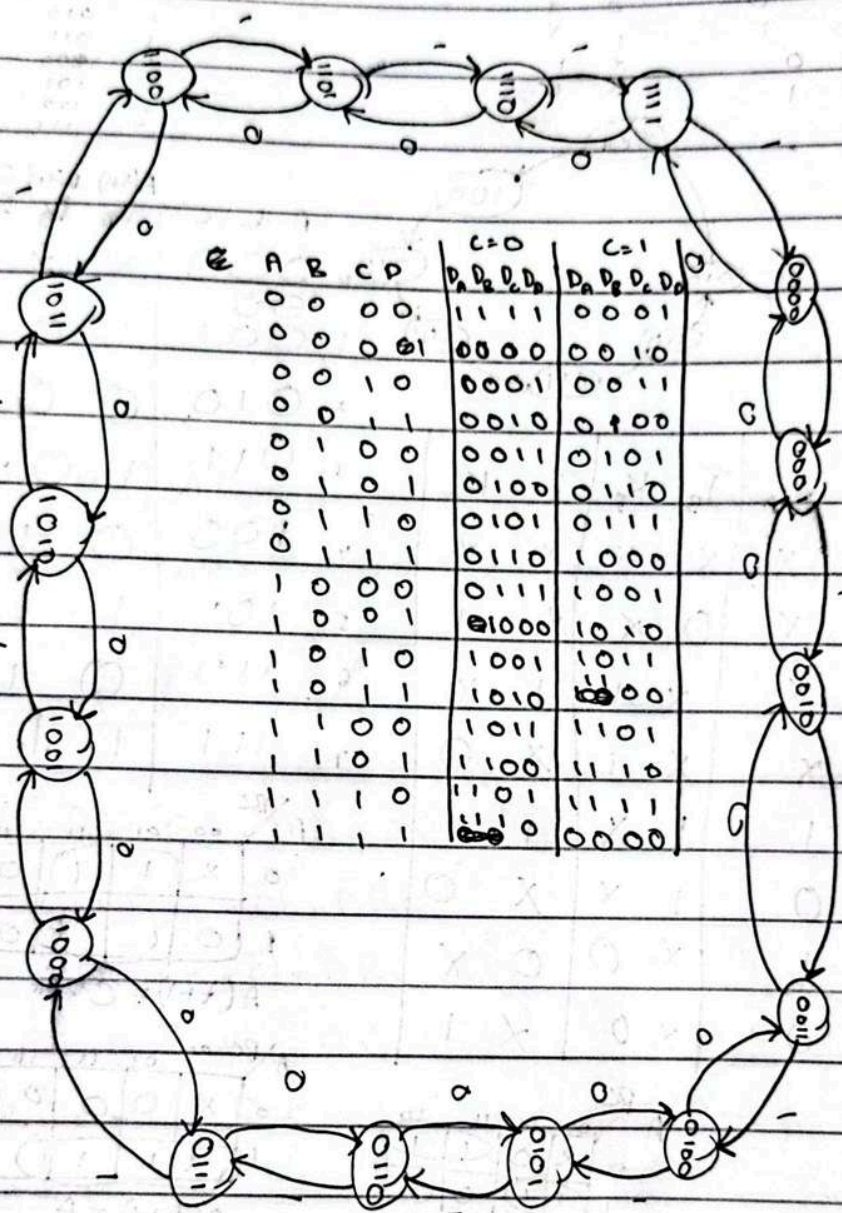
A \ BC	00	01	11	10
0	X	0	0	0
1	1	1	1	1

$$B(t+1) = A$$

A \ BC	00	01	11	10
0	X	0	1	1
1	1	1	0	0

$$C(t+1) = A\bar{B} + \bar{A}B$$

$$= A \oplus B$$



C	A	B	C	D	C=0				C=1			
					D _n	D _{n-1}	D _{n-2}	D _{n-3}	D _n	D _{n-1}	D _{n-2}	D _{n-3}
0	0	0	0	0	1	1	1	1	0	0	0	0
0	0	0	0	1	0	0	0	0	0	0	1	0
0	0	0	1	0	0	0	0	1	0	0	1	1
0	0	0	1	1	0	0	1	0	0	1	0	0
0	0	1	0	0	0	0	1	1	0	1	0	1
0	0	1	0	1	0	1	0	0	0	1	1	0
0	0	1	1	0	0	1	0	1	0	1	1	1
0	0	1	1	1	0	1	1	0	1	0	0	0
1	0	0	0	0	0	1	1	1	1	0	0	1
1	0	0	0	1	0	1	0	0	1	0	1	0
1	0	1	0	0	1	0	0	1	1	0	1	1
1	0	1	0	1	1	0	1	0	1	0	0	0
1	1	0	0	0	1	0	1	1	1	1	0	1
1	1	0	0	1	1	0	0	0	1	1	1	0
1	1	1	0	0	1	1	0	1	1	1	1	1
1	1	1	0	1	1	1	0	0	0	0	0	0

0	1	0	0	1	1	1	0	0
0	1	0	1	1	1	1	1	0
0	1	1	0	1	1	1	1	1
1	0	0	0	0	0	1	0	0
1	0	0	1	0	0	1	0	0
1	0	1	0	1	1	0	0	0
1	1	0	0	1	1	0	1	0
1	1	0	1	1	1	0	1	0
1	1	1	0	1	1	0	1	0
1	1	1	1	1	1	0	1	0

$$\begin{matrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{matrix}$$

$$\bar{A}\bar{B} + AB$$

$$1 + 0$$

$$1$$